

MascotRL

Auditable Trading Personalities in Portfolio Allocation via Reinforcement Learning

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Abstract

Regulators demand a name. UCITS mandates, Form N-1A disclosures, and the SEC Names Rule all require that a fund carry a strategy label verifiable against its holdings: momentum fund, contrarian fund, risk-parity fund. The label lets compliance officers check for mandate drift, lets investors match a product to a portfolio need, and lets regulators trace losses to stated risk appetite. Yet the neural equity allocator trained by reinforcement learning offers only a path of weights and a performance number; it does not announce whether it chases winners, bets on reversals, hedges volatility, or rotates across sectors. MascotRL closes this gap by reading the allocator as one animal body, a zoomorphic anatomy that maps each design choice to an organ: muzzle for the lagged market state that enters the network, neck for the layers that transform it, heart for the reward signal that steers learning, haunch for the simplex head that shapes weight geometry, tail for the transaction cost that penalises churn, and spine for the policy-gradient update rule that tunes parameters toward higher cumulative reward. From this anatomy, a locked scorecard assigns each book that clears an active-share screen to one of six trading personalities before the out-of-sample census runs, ensuring the classification cannot be tuned to flatter the results: trend follower for momentum-loading books, contrarian for reversal-loading books, risk manager for variance-minimising books, speculator for lottery-tilted books, tactical rotator for regime-switching books, and generalist for blended books that span multiple styles. The screen excludes closet indexers by withholding the name when mean active share versus the equal-weight baseline falls below 12.5%, a threshold that demands genuine deviation from passive exposure. On the 179 books that clear this screen, the scorecard files 11 trend followers, 32 contrarians, 18 risk managers, 21 speculators, 19 tactical rotators, and 78 generalists, each recorded with its runner-up personality and the margin of victory.

Keywords: MascotRL; explainable reinforcement learning; portfolio allocation; policy-level explanation; trading personality; transaction costs; allocation-head geometry; style analysis; model risk.

JEL: G11, G14, C45, C61.

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Notation

The letter z is used for three objects. z_{t-1} is the lagged feature vector in the state. z_t is the vector of name scores leaving the torso. $z(\cdot)$ is the standardisation of a behaviour coordinate on the evaluation panel.

Table 1. Core notation.

Symbol	Meaning	Domain
State and book		
t	Decision date	trading day
i	Slot in the book	$1, \dots, K$
K	Number of slots. The census uses 100 or 200. Three 400-name books sit outside the style matrix	100 or 200
z_{t-1}	Lagged feature vector: implied variance, term slope, skew, and related point-in-time channels	equity and option panel
w_{t-1}	Holdings entering the step	Δ^{K-1}
s_t	State: $(z_{t-1},\ w_{t-1})$, together with days held, cumulative cost, and the equal-weight base	Eq. (2)
$a_t = w_t$	Portfolio weights sent to the market	Δ^{K-1}
Δ^{K-1}	Long-only unit simplex, $\{w \geq 0 : \sum_i w_i = 1\}$	feasible books
π_θ	Policy, a map from the state to a distribution over books	$s \mapsto$ books
θ	Parameters of π	network weights
z_t	Logits, one score per slot, before the allocation head	$\mathbb{R}^K$
Reward, cost, and update		
r_t	Reward after the rebalance charge	$\mathbb{R}$
c_t	Rebalance charge inside the step, $\kappa \sum_i \lvert w_{t,i} - \tilde{w}_{t-1,i} \rvert$	Eq. (4)
κ	Proportional cost on the full weight change, charged in training and in evaluation	5 basis points
$\tilde{w}_{t-1}$	Previous book after price drift	Δ^{K-1}
γ	Discount on future reward	0.99
λ	Decay in generalised advantage estimation	0.95
ε	Clip on the PPO probability ratio	0.3
Holdings and scorecard		
AS_t	Active share versus the equal-weight book $1/K$	$\frac{1}{2} \sum_i \lvert w_{t,i} - 1/K \rvert$
AS	Active share on the stored path	$\mathrm{median}_t(\mathrm{AS}_t)$
$L_1(w, 1/K)$	Absolute gap versus equal weight, equal to 2 AS. The screen withholds a name when the mean gap is below 0.25	$[0,\ 2 - 2/K]$
HHI	Concentration, $\sum_i w_{t,i}^2$	$[1/K,\ 1]$
One-way turnover	Half the L_1 change in weights. The holding period on the scorecard is its reciprocal	$[0,\ 1]$
$z(\cdot)$	Behaviour coordinate standardised on the evaluation panel	scorecard input
S_{trend}, $S_{\text{contrarian}}$, S_{risk}, $S_{\text{speculator}}$, S_{rotator}	Frozen personality scores. The winning score files the name when it leads by 0.25; otherwise the book is a generalist	real numbers

1 Introduction

An investor chooses a portfolio under risk. Each day the book is a non-negative split of one unit that sums to one. A committee-readable book is usually a stack of data, an alpha rule, risk limits, costs, and portfolio construction. Here most of that stack is one map, and the head is what still sits between scores and a book. A supervisor can check the mandate of that portfolio only when the book has a name a case officer can read. A reinforcement-learning equity allocator writes a path of holdings. That path does not, by itself, state the strategy.

Option prices summarise what the market is currently pricing about later moves. A sequential allocator has to turn those lagged signals into weights, and the same cost is charged when the book is trained and when it is read. A sorted portfolio never faces that charge.

> *Can the behaviour of a reinforcement-learning equity allocator be summarised by a pre-registered trading personality that a supervisor can verify from holdings alone, with the same evidence used to approve a conventional strategy's stated investment policy?*

We file each book that clears a closet-index screen as one of six personalities. The screen (active share against equal weight) withholds the name when the mean absolute gap versus equal weight is below one quarter of the fund. The scorecard that files the name was locked before the census. On the 179 books that clear the screen, it files 11 trend followers, 32 contrarians, 18 risk managers, 21 speculators, 19 tactical rotators, and 78 generalists. The personality card records the label, the runner-up, the gap, and the screen.

The same book is then opened as one body. The muzzle is the lagged state, the neck is the network torso, the heart is the reward, the haunch is the allocation head, and the tail is the rebalance charge. Sparsemax can leave a weight at exactly zero. Softmax cannot. No tested book is offered as a beat of a same-cost benchmark.

Section 2 maps what a case officer already asks for. Section 3 places the paper in the literature. Section 4 defines the scorecard. Section 5 describes the design and reports the census. Section 6 opens one body for each personality. Section 7 reverse-maps each personality. Section 8 records what the charge costs. Section 9 shows the personality card. Sections 10 and 11 close.

2 Institutional setting: why opaque allocators stall

2.1 How a strategy reaches investors today

In the European Union a UCITS reaches investors only after a national competent authority approves constitutive documents and a prospectus that states objectives, strategies, limits, and risks (Commission de Surveillance du Secteur Financier, 2025). Luxembourg's CSSF recommends an explanatory memo for complex strategies (Commission de Surveillance du Secteur Financier, 2026). Processing times for that memo, and for a standard Irish file, are in Appendix D.

In the United States a registered fund discloses principal investment strategies on Form N-1A. The Names Rule, as amended in 2023, requires an 80% investment policy for funds whose names suggest particular characteristics, with quarterly review and a ninety-day cure; compliance dates run from June or December 2026 depending on fund-group size (U.S. Securities and Exchange Commission, 2023; U.S. Securities and Exchange Commission, 2025a). Advisers face Form ADV disclosure and the Marketing Rule's substantiation standard.

2.2 Where algorithms add gates

When the strategy is algorithmic, MiFID II Article 17 requires notification to the home competent authority and, on request, a description of the nature of the strategies, parameters, controls, and testing (European Parliament and Council, 2014). RTS 6 adds an annual self-assessment and validation report (European Commission, 2017). The FCA's MAR 7A implements the same request with a fourteen-day clock.

Model-risk regimes tighten the same demand. SR 11-7 asks for conceptual soundness and effective challenge (Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency, 2011). Canada's OSFI Guideline E-23 (final September 2025, effective May 2027) explicitly flags "black box approaches" as needing alternative controls and ties explainability to purpose, autonomy, and stakeholder impact (Office of the Superintendent of Financial Institutions, 2025). BaFin's 2021 papers on big-data AI and machine learning in risk models make the same point for German supervised firms (Bundesanstalt für Finanzdienstleistungsaufsicht, 2021a; Bundesanstalt für Finanzdienstleistungsaufsicht, 2021b).

2.3 AI-specific expectations

ESMA's May 2024 statement on AI in retail investment services, IOSCO's 2021 and 2026 reports, and the EU AI Act (portfolio management is not Annex III high-risk, so sectoral law governs) all push toward governance, oversight, and documentation rather than a new product licence (European Securities and Markets Authority, 2024; International Organization of Securities Commissions, 2021; International Organization of Securities Commissions, 2026; European Parliament and Council, 2024). The FCA's April 2024 AI Update says existing rules, including SM&CR and the Consumer Duty, already apply (Financial Conduct Authority, 2024). MAS FEAT principles 12 to 14 ask for clear explanations of data and consequences "upon request," and note that clear explanations "do not necessitate exposure of intellectual property or publishing of proprietary source codes" (Monetary Authority of Singapore, 2018); the 2025 AI risk-management consultation extends lifecycle controls to fund management (Monetary Authority of Singapore, 2025). Hong Kong's SFC circular 24EC55 treats investment advice and research as high-risk GenAI use cases that need validation before use (Securities and Futures Commission, 2024). FINRA Notice 24-09 applies existing supervision rules to AI tools (Financial Industry Regulatory Authority, 2024). The SEC withdrew its predictive-data-analytics conflict proposal in June 2025 (U.S. Securities and Exchange Commission, 2025b), but March 2024 AI-washing orders against Delphia and Global Predictions show that Marketing Rule substantiation already bites (U.S. Securities and Exchange Commission, 2024).

2.4 Where the black box becomes the thorn

Four friction points recur. First, there is no prose strategy description that a case officer can put next to a KIID or N-1A narrative. Second, label-holdings consistency is already supervised: ESMA's 2016 closet indexing review flagged 5 to 15 percent of about 2,600 UCITS equity funds (European Securities and Markets Authority, 2016), and the Names Rule does the same job in the US. Third, validation and effective challenge require a conceptual story, not only a backtest. Fourth, marketing claims about AI must be substantiable.

2.5 What a trading personality supplies

Table 2 maps those requests to the evidence in this paper. Narang separates the alpha model from the execution layer (Narang, 2013). The personality card reads the holdings rather than a fill. The personality card records a filing a case officer can read next to a mandate. The card is the annex set next to a prospectus. It does not replace the prospectus.

Table 2. Supervisory request and the MascotRL evidence that answers it.

Regime	Request	MascotRL evidence
UCITS / N-1A / ADV	Strategy description	Personality card + mascot definition
Names Rule / closet indexing	Label-holdings consistency	Frozen scorecard + active share
Ongoing supervision	Drift monitoring	Label change, or 63-day active share below 12.5 percent
SR 11-7 / OSFI E-23	Conceptual soundness	Anatomy (Section 6)
Validation practice	Validation	Purged combinatorial CV + frozen scorecard
Marketing Rule / FEAT	Substantiation	Fail-closed claim tiers

3 Related literature

Moody and Saffell (2001) train a trader by ascending an online differential Sharpe ratio. Jiang, Xu, and Liang (2017) show that a deep net can output a long-only book with a commission inside wealth. This study keeps that score and asks whether the map from scores to weights moves the holdings far enough from equal weight to name a style.

Martins and Astudillo (2016) show that sparsemax can place exact zeros. Peters, Niculae, and Martins (2019) place 1.5-entmax between softmax and that projection. Jiang, Xu, and Liang (2017) also note that softmax makes the book long-only.

Sharpe (1992) reads style from a return regression, not from holdings. Daniel et al. (1997) read holdings against characteristic benchmarks. Cremers and Petajisto (2009) define active share as half the absolute gap versus a benchmark. Here the benchmark is equal weight, and the gap only decides whether a name is visible.

DeMiguel, Garlappi, and Uppal (2009) find that optimised rules often fail to beat one-over-N. That contest is a boundary. It is not this paper's finding.

4 The MascotRL framework

4.1 Six personalities and their human analogues

Each personality has a human peer used later for style-matched comparisons: twelve-minus-one-month momentum for the trend follower (Moskowitz, Ooi, and Pedersen, 2012), one-month reversal for the contrarian (Jegadeesh, 1990), shrunk minimum variance for the risk manager (Clarke, Silva, and Thorley, 2006), the MAX lottery for the speculator (Bali, Cakici, and Whitelaw, 2011), volatility-managed equity for the tactical rotator (Moreira and Muir, 2017), and a blend for the generalist.

4.2 Behaviour vector and frozen scorecard

The style matrix columns used for this census are residual trend, reversal, defensive, and lottery. From the holdings the scorecard builds a behaviour vector, z-scores each coordinate on the evaluation panel, and dots the vector with the frozen weights. A name is filed only if the winner leads by 0.25. Otherwise the book is a generalist. The scorecard was locked before the census. The reward charged inside the step is the estimand of that environment (Hilpisch, 2024). The behaviour vector reads the weights after that charge. A change of reward would be a different estimand.



Trend follower
Cheetah
$n = 11$



Contrarian
Fox
$n = 32$



Risk manager
Tortoise
$n = 18$



Speculator
Magpie
$n = 21$



Tactical rotator
Hummingbird
$n = 19$



Generalist
Owl
$n = 78$

Figure 1. Six trading personalities. The animal nickname sits under the name. Counts are the books that clear the closet-index screen.

4.3 Active share as the visibility gate

Following Cremers and Petajisto (2009), define active share against the universe's $1/K$ portfolio as

$$\mathrm{AS}_t = \tfrac{1}{2}\sum_{i=1}^{K}\left|w_{t,i} - \tfrac{1}{K}\right|, \qquad \mathrm{AS} = \mathrm{median}_t(\mathrm{AS}_t). \tag{1}$$

Equivalently, $\mathrm{AS} = \frac{1}{2}L_1(w, 1/K)$. The closet-index screen withholds a style label when the mean absolute gap versus equal weight is below 0.25, which is an active share of 12.5 percent. Call that gap the visibility gate. The rule uses holdings only. A large gap from equal weight is a difference, not evidence of skill.

4.4 What the card records

The personality card records the label, the runner-up, the gap, and the screen. A label change, or a 63-day active share below 12.5 percent, is a drift trigger. The claim tier on the card is interpretation only. The card is a reading of the weight path. It does not replace the validation report a supervisor already asks for.

5 Data and experimental design

5.1 Universe

The investable set is built from point-in-time S&P 500 constituents. Those lagged channels are standardised with moments frozen at the previous close. A missing print is filled from the last available past value. A feature dated after the decision is not part of that state. The licensed sources are OptionMetrics, CRSP, and Refinitiv. Sector labels are a GICS proxy. No option is traded. At each monthly rebalance the substrate selects $K \in \{100, 200\}$ dynamic slots by a correlation-cluster rule, so column i is a slot, not a fixed firm. Table 3 summarises GICS sector shares of slot-days. The ten most frequently held firms in the $K = 100$ map include Molson Coors (TAP), Constellation Brands (STZ), Newmont (NEM), Super Micro (SMCI), and First Solar (FSLR); the full list appears in Appendix B.

Table 3. GICS sector composition of slot-days ($K = 100$).

Sector	Share of slot-days (%)
Health Care	16.5
Information Technology	15.6
Consumer Discretionary	14.6
Consumer Staples	13.0
Industrials	11.4
Communication Services	8.1
Financials	7.5
Materials	5.2
Utilities	2.9
Real Estate	2.7
Energy	2.1

5.2 Experiment grid

Allocators vary by algorithm (PPO, DDPG, SAC, TD3, CVaR-constrained PPO, Monte Carlo policy gradient, direct reinforcement learning, DQN), network torso (MLP, GRU, LSTM, transformer), allocation head (softmax, sparsemax, bounded tanh, 1.5-entmax, Dirichlet), reward (mean-variance, mark-to-market P&L, differential Sharpe, CVaR, and related objectives), and universe size. Training uses GELU activations, discount $\gamma = 0.99$, GAE $\lambda = 0.95$, and PPO clip $\varepsilon = 0.3$ where applicable. Friction of five basis points of turnover sits inside every environment step. The same charge is taken while the policy is trained, so the book is not learned in a cost-free world and then scored with costs.

Parameter	On-policy (PPO / constrained PPO)	Off-policy (DDPG / SAC / TD3)	Notes
Network	MLP, often two hidden layers	same family	recurrent variants also tested
Activation	GELU	GELU	matches the training configuration
Discount γ	fixed per run	fixed per run	part of the estimand
Clip ratio	0.3 (from the training configuration)	not used	on-policy only
GAE λ	0.95 (typical)	not used	advantage baseline
Entropy bonus	small positive	SAC entropy target	exploration has a dollar cost
Tail constraint	CVaR overlay on constrained PPO	not used	Lagrangian form
Transaction cost	inside each step	inside each step	5 bp of $\lvert\Delta w\rvert$

5.3 Out-of-sample protocol and sample rule

Evaluation uses purged combinatorial cross-validation (CPCV) (López de Prado, 2018): many train and test splits, with a gap at each split so a label near the boundary cannot leak into training. It is not walk-forward. Nested walk-forward is a different procedure, and this paper does not relabel one as the other. The splits follow the date order. Days are not shuffled and then treated as separate draws. A path can still be memory of the sample or a feature dated after the decision. The purged splits exist so those two are not treated as evidence. The style matrix covers the finished hundred-slot books. Three 400-name books are outside it (Appendix C).

Table 4. Personality census on the books that clear the closet-index screen. A book inside the screen is not given a style name.

Personality	Books	Share (%)
Trend follower	11	6.1
Contrarian	32	17.9
Risk manager	18	10.1
Speculator	21	11.7
Tactical rotator	19	10.6
Generalist	78	43.6
Clear the screen	179	100

Table 5. Allocation head on sixty books: one proximal-policy network and a plain feed-forward torso, three heads, and four rewards (mean-variance, mark-to-market, a root semi-quadratic penalty, and a semi-mean-squared error), seeds 0 through 4. Concentration is the median across books of mean Herfindahl on the stored holdings path. Turnover is the median of one-way turnover on that path, half the L1 change in weights. The five-path Sharpe column is the median of the mean of five cross-validation path Sharpes. It is context.

Head	Books	Concentration	Turnover	Holdings entropy	Five-path Sharpe
Dense	20	0.040	0.068	4.29	0.77
Sparse	20	0.413	0.443	1.40	0.05
Bounded	20	0.011	0.047	2.25	-1.06

Eight contrarian books use a 1,485-day calendar from 4 January 2019 and omit 24, 27, and 28 February 2020 and 9, 12, 16, and 23 March 2020. They stay in the holdings census and out of any performance sentence.

Figure 2 is the architecture the rest of the paper explains.

5.4 Census

The six counts sum to 179. A book inside the closet-index screen is not in the table.

5.5 Designed versus observed

On the 106 books that were already named, a strict designed-versus-observed match occurs 7 times. The comparison is between the name the design carried and the name the scorecard filed, on the same costed books. The scorecard's holding period and its turnover are one clock. The holding period is the reciprocal of one-way turnover, and that identity holds on the 105 stored books. The match is rare. Twenty-five of these 106 books share the designed label and the filed label. The strict match of 7 is a tighter rule than that shared label.

5.6 Allocation head

On 60 books, 20 for each head, the dense head has concentration 0.040 and turnover 0.068, the sparse head 0.413 and 0.443, and the bounded head 0.011 and 0.047. The five-path Sharpe ratios in the same table are context, not the finding.

Matched-head twins, with their own denominator, are in Appendix E.

6 Anatomy of a personality

A quantitative book is an alpha model, a risk model, and a transaction cost model, brought together by portfolio construction and then handed to execution (Narang, 2013). A reinforcement-learning allocator has the same

organs under different names. This section follows one trend-follower book. The census counts 179 books, of which 11 are trend followers.

The reading is zoomorphic: each part of the allocator is named after a body part, and each personality is named after an animal. The mascots are labels for holdings, not species the book was trained to become.

A cheetah, a fox, a tortoise, a magpie, a hummingbird, and an owl share one body. The muzzle takes in lagged features and the previous weights. The neck turns that observation into a score for each name. Three gold ticks at the collar are the fixed map onto the simplex: softmax leaves every name a crumb, sparsemax can leave a name empty, and bounded tanh caps each score and then fills the body again. The heart is the reward. The haunch is the allocation head, the step that becomes weights. The tail is the cost inside the step, charged in training and in evaluation. The spine is the learning update: it carries the reward back so the next day's map changes. If that cost were charged only after training, the tail would not be on the path the haunch just took. The species is not in these parts. The species is the name the scorecard files from the holdings. The loop is trial and error with a scorekeeper. The agent sees a lagged observation, proposes weights, and is graded by a reward.

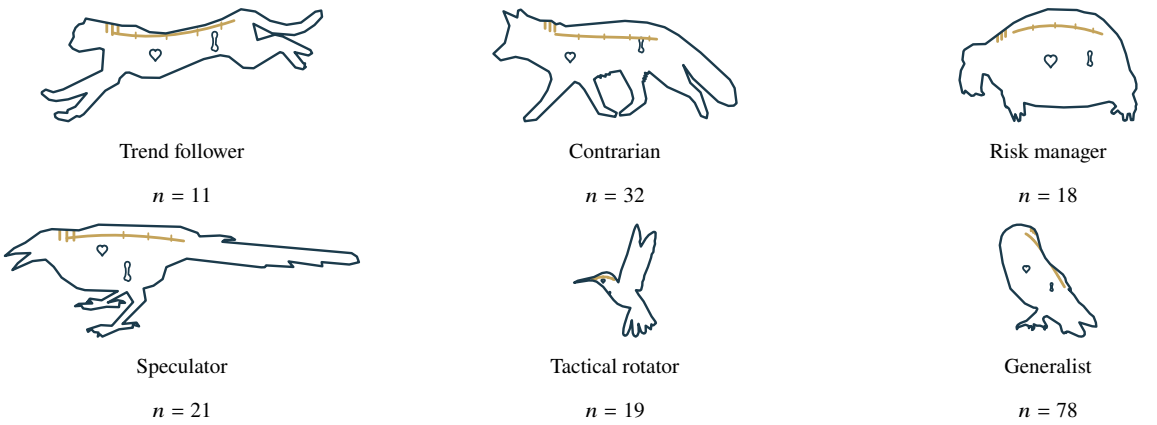


Figure 6. The same marks on six skins. The personality is the name the scorecard files from the holdings. CC0 silhouettes from PhyloPic (Michaud; Sayol; Mauel; Jablunovsky; Sayol; Sayol).

None of the six pictured books beats the equal-weight book on that same charge.

The plate names those parts in the trainer. The state is the lagged observation the actor conditions on (Plaat, 2022). The actor proposes the next weights. The critic estimates the value of that state. The advantage is how far the after-cost outcome sits from that estimate, and the update weights the proposal by that gap (Sutton and Barto, 2018; Zai and Brown, 2020). The rebalance charge is inside the outcome the critic sees, in training and in evaluation (Hilpisch, 2024). A differential Sharpe score and mark-to-market profit are two such outcomes (Moody and Saffell, 2001).

6.1 Market state

At date t the agent observes a point-in-time feature vector z_{t-1} assembled from the equity panel and option-implied channels, together with previous holdings w_{t-1}:

$$s_t = (z_{t-1},\, w_{t-1}). \tag{2}$$

The lagged feature vector in that state is 30-day model-free implied variance, the implied-volatility term slope, and 30-day skew. The same state carries the previous weight, days held, cumulative cost, and the equal-weight

base. The muzzle is this state, lagged features and the previous holdings taken together. These inputs are observations. The list does not prove that the state is Markov-complete (Plaat, 2022). The muzzle registers what has already arrived: lagged features, not the latent state behind them. The market can move, so the same lagged observation is not a promise of the same next-day transition (Plaat, 2022). A feature dated after the decision is not part of that intake. The previous weights are the holding already sent, and a later print cannot enter the step already taken. Previous weights matter because costs make the control problem path-dependent.

6.2 Torso

The policy maps a state to a portfolio, or to a distribution over portfolios (Zai and Brown, 2020).

An MLP with GELU hidden layers maps the flat state to K logits (and a scalar value head):

$$h = \mathrm{GELU}(W_2\,\mathrm{GELU}(W_1 s_t + b_1) + b_2), \qquad z_t = W_z h + b_z. \tag{3}$$

The MLP keeps no hidden state past today's observation. A GRU or LSTM torso does not by itself file a style, because the head still maps scores to weights. The neck is this layer, the last step before each name has a score. Where the trainer is an actor-critic method, the actor writes the weights and the critic scores them (Zai and Brown, 2020). The neck joins those two maps. They are not one head. An MLP keeps a tilt only when today's lagged features still carry it. A recurrent torso can hold a fading trend across days. That is a hypothesis, not a fact assumed up front. The risk-manager book in this paper is the LSTM case named below. The policy here is a strategy in the portfolio-management sense: a mapping from information to a proposed book.

6.3 Allocation head

The action sent to the market is the vector of portfolio weights (Jiang, Xu, and Liang, 2017). A Dirichlet policy places weight on the simplex, so the budget and the long-only constraint hold by construction. It does not cap any one name (Xue, Ye, et al., 2025).

Logits become simplex weights. Softmax never places exact zeros; sparsemax is the Euclidean projection onto the simplex and can zero names (Martins and Astudillo, 2016); bounded tanh clips then renormalises; a Dirichlet mean stays interior. The haunch is this head: it turns each name's score into a weight, and a sparsemax head can leave some weights at zero (Martins and Astudillo, 2016). The haunch is that step. The weights are the action (Plaat, 2022). A projection onto the simplex keeps the weights feasible. It is not a risk model. That feasibility is not a trading style (Jiang, Xu, and Liang, 2017). The collar in the drawing is that fixed map. Scores may move, and the collar forces them onto the simplex before a weight exists. Exploration noise is applied before the collar, and the collar then forces the scores onto the simplex (Zai and Brown, 2020). The collar can shrink that noise. Variety in the scores is then not variety in the holdings (Zai and Brown, 2020). On the 60-book PPO MLP factorial, the heart can change without moving the haunch. Inside a fixed head, the median one-way turnover moves by about 0.001 for softmax, 0.004 for bounded tanh, and 0.066 for sparse. The heads themselves sit near 0.047, 0.068, and 0.43. Lag-1 weight autocorrelation stays near 0.92, 0.60, and 0.41. Changing the reward inside one head is not a change of the head. The head is what separates those habits.

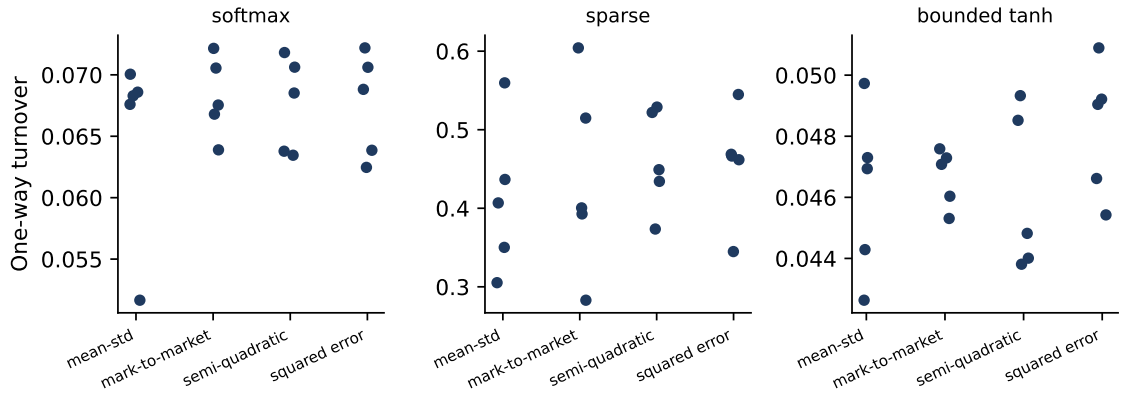


Figure 7. One-way turnover for five seeds at each reward, with the allocation head held fixed. Each panel is one head.

The three-name picture below is the same fact on 3 July 2019. The dense head stays inside the triangle, near one third on each of the three largest names. The sparse head sits on an edge, because one of those three weights is exactly zero and the other two still sum to one. A softmax head cannot make that edge. On the stored paths, a sparse head parks most names at an exact zero. A softmax head and a bounded tanh head almost never do.

A book can still be concentrated with a softmax head, if the scores that enter the head are already peaked. The speculator sheet is that case: the head is softmax, and the concentration of the stored weights is 0.99. Grinold and Kahn (2020) write the transfer coefficient as the correlation between the ideal paper portfolio and the portfolio actually held. The allocation head is that transfer coefficient in software: it decides how much of the network's conviction reaches the book.

On the 60-book table the dense head's concentration is 0.040, which is above the equal-weight floor, and the bounded head at 0.011 is the low-concentration case.

6.4 Friction inside the step

Equity rebalance cost is

$$c_t = \kappa \sum_{i=1}^{K} \left| w_{t,i} - \tilde{w}_{t-1,i} \right|, \qquad \kappa = 5\,\text{bp}, \tag{4}$$

with $\tilde{w}$ the drifted previous book. The tail is this charge, the configured rate on the change in weights, the same charge in training and in evaluation. Research cost is commission, spread, and impact, each stated on its own (Narang, 2013). Equation 4 is one linear charge on the weight change. An impact curve is a different object. If that charge were applied only after training, the learned book would not be the book that is read. Among the six pictured books, stored turnover is 0.27 where the weights keep moving, and 0.025, 0.010, 0.006, 0.002, and 0.001 where the book stays. The outline names that habit. That charge is the configured rate. The stored cost per unit of the full change in weights is higher, and it differs by head. On 105 books the median is about 7.0 basis points for 1.5-entmax, 6.5 for the sparse head, 10.2 for softmax, and 13.4 for bounded tanh. A five-basis-point charge on that full change would have sat near 5. A few softmax and bounded-tanh books sit near 120 basis points. They are not the center of the bars. The behaviour card reports one-way turnover, about half the full change, so the same cost per unit of the card's turnover is about twice the figure.

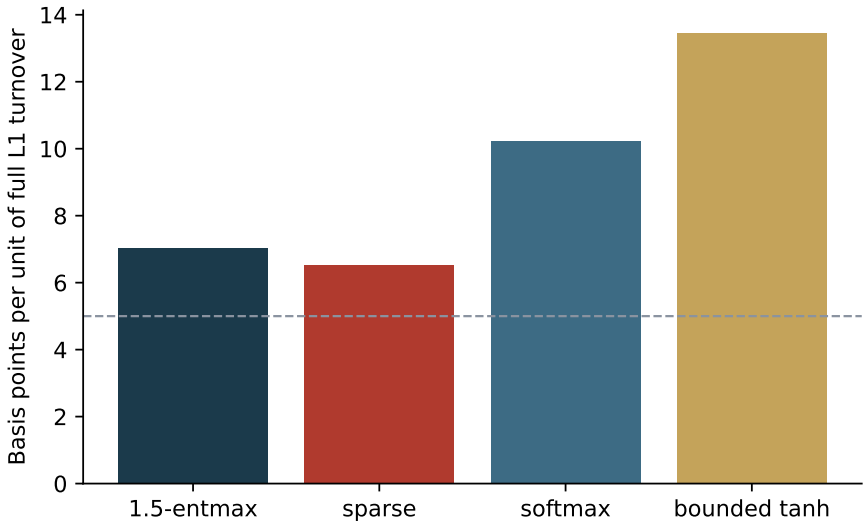


Figure 10. Stored cost divided by the full sum of absolute weight changes, median by allocation head. The horizontal line is five basis points.

Once a charge sits on every rebalance, a tiny change in weights is often not worth trading. Exploration that ignores costs invents false policies. The tail loads each step as it is taken, including the exploratory ones, so the book that is trained is the book that is read (Plaat, 2022).

6.5 Reward

Dense step rewards are after-cost residual returns. The Trend exemplar uses a squared-error hedging episode objective; other books use mean-variance, differential Sharpe (Moody and Saffell, 2001), or CVaR. A differential Sharpe score adjusts the path for recent risk. Mark-to-market profit is a different outcome (Moody and Saffell, 2001). The reward is the mandate letter written in mathematics. That mandate pays for one habit, such as a squared-error hedge, raw profit, a tail loss, or compounding per unit of recent volatility. The heart is this reward, the mandate the book is scored against after costs. Changing the reward without locking a new estimand would mix books that were not scored on the same outcome. The reward is the estimand of the environment (Hilpisch, 2024). A silent change of that reward is a different book. An extra term laid on top of that named objective would be a different book (Plaat, 2022). A textbook condition for a harmless extra term is not a condition this census uses.

6.6 Learning rule

Actor-critic methods learn a policy and a value function (Sutton and Barto, 2018).

Discounting uses $\gamma = 0.99$. Changing the discount of 0.99 would retarget the learning update toward a shorter or a longer horizon (Plaat, 2022). Advantages use GAE with $\lambda = 0.95$. The choice of advantage estimator is a bias-variance decision, and this census keeps generalized advantage estimation fixed (Plaat, 2022). PPO clips the probability ratio at $\varepsilon = 0.3$. DDPG, TD3, and SAC, which the experiment grid already lists, score proposed books on a replay of past days. The discount of 0.99 lets a drawdown that lasts several weeks still enter the update. That replay sits off the policy being updated, unless it is reweighted back onto that policy (Plaat, 2022). The reweighting does not remove drift in the panel. A baseline, or an advantage estimator, reduces the noise in that update. It does not repair a reward that measures the wrong outcome (Plaat, 2022). The spine in the drawing is this update: an advantage-weighted step on the current batch (Zai and Brown, 2020). PPO reinforces actions that raised the score, and the clip keeps one update from jumping the map.

A second seed barely moves a bounded-tanh book or a softmax book. It can move a sparse book by a wide margin: in the widest mark-to-market family the Herfindahl runs from 0.279 to 0.657. The entropy in the training log is not the entropy of the holdings, and it does not separate heads. The update does not file the personality. It changes the mapping from the lagged state to the next book, using the reward after the charge. PPO keeps the new mapping close to the previous one by clipping the probability ratio. DDPG and DQN score a proposed book on a replay of past days and then move the mapping. What the census later reads is the holdings that mapping produced, not the name of the algorithm. The six pictured books report concentration and turnover on the stored path. Those two series do not separate the torso, the reward, and the head, so this section does not fit a regression of the personality on the parts.

6.7 From weights to personality

Economic distillation projects a complex model onto a linear model or into words so the drivers can be read (Cong et al., 2021).

Behaviour measures are z-scored, dotted with frozen scorecard weights, and labelled if the margin is at least 0.25; otherwise Generalist. The personality is the distillation of the policy into the language of a mandate. The six outlines below keep the five marks from the opened body and change only the skin. The skin is the name the scorecard filed from the holdings.

Reviewer checklist. A reviewer can verify: (1) every feature channel is lagged and point-in-time, (2) architecture and activation match the training configuration, (3) the allocation head matches the stored weights, (4) costs are identical in training and evaluation, (5) the reward name matches the configuration, (6) learning hyperparameters and seed are recorded, and (7) the primary score, runner-up, and margin appear on the personality card.

6.8 Trend follower

On the stored holdings path, the sparsemax head and the squared-error hedging reward leave concentration 0.58 and turnover 0.27. The cheetah is that pace: the book keeps moving. The marks are the muzzle, the neck, the heart, the haunch, and the tail, the same parts as on the opened body.

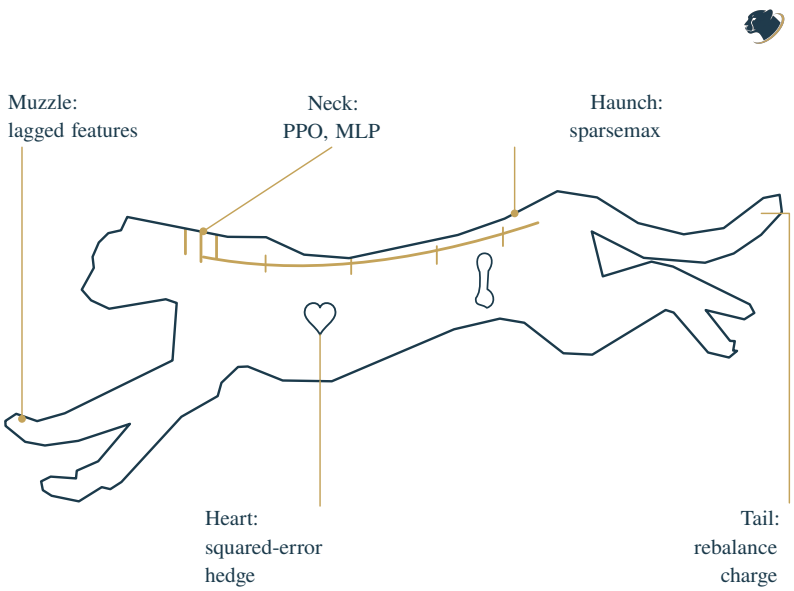


Figure 11. Trend follower (Cheetah). The marks are the parts of the opened body. Turnover 0.27, a book that keeps moving.

6.9 Risk manager

On the stored holdings path, the sparsemax head, the LSTM torso, and the mean-standard-deviation reward leave concentration 0.46 and turnover 0.010. The shell is a book that stays.

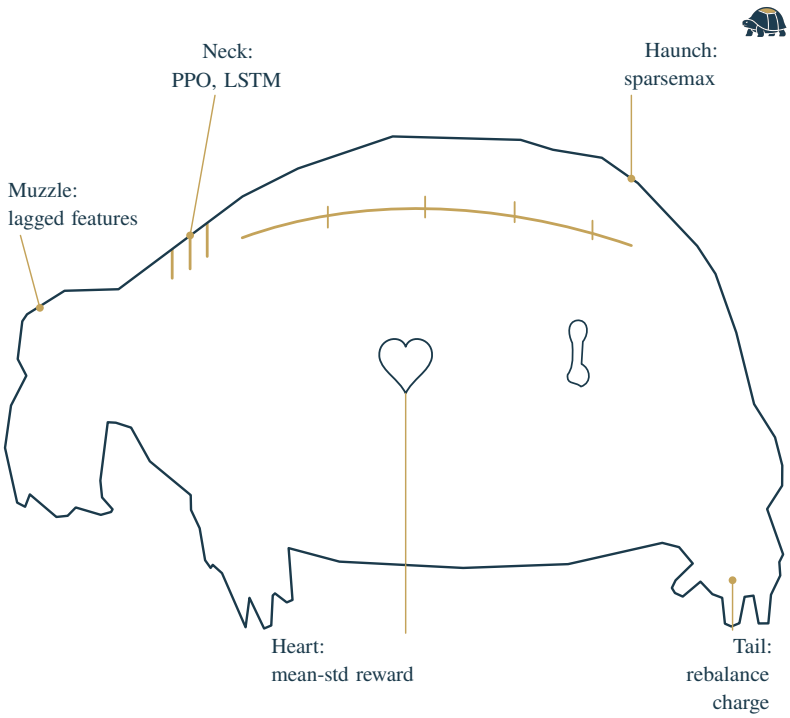


Figure 12. Risk manager (Tortoise). The shell marks a book that stays. Turnover 0.010.

The Tortoise's largest weight sits near 0.59. One-way turnover is 0.010, so a zero, once set, is held, and concentration on the stored path is 0.46. The book is PPO with an LSTM torso, a sparsemax head, and a mean-standard-deviation reward on 200 names. A long-only minimum-variance desk also holds a short list, and Clarke, de Silva, and Thorley cap any one name at 3 percent (Clarke, Silva, and Thorley, 2006). This book's largest weight is far above that cap. Its market loading is 0.99, so the quiet book still moves with the market.

6.10 Speculator

On the stored holdings path, the softmax head and the mark-to-market reward leave concentration 0.99 and turnover 0.002. The outline is a book gathered onto a few names.

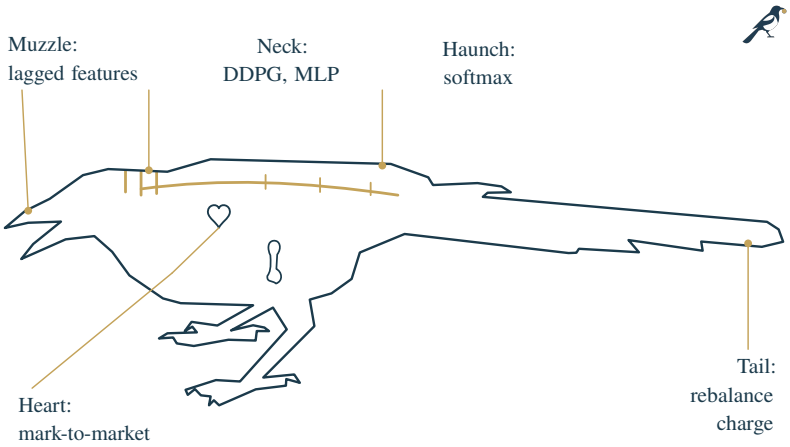


Figure 13. Speculator (Magpie). Concentration 0.99, a book gathered onto a few names.

The pictured Magpie is a softmax hold with concentration 0.99 and turnover 0.002. On one stored day that hold is a single slot at weight 1, and that day’s turnover is zero. The census files this hold as a generalist. The book the census files as a speculator is a different one: a GRU sparsemax book, concentration 0.68 and turnover 0.020. A lottery desk buys the names with the largest recent daily move (Bali, Cakici, and Whitelaw, 2011). What matches that desk in the stored row is the concentration, and the profitability loading of −0.19. The firms in that GRU book are not named here.

6.11 Tactical rotator

On the stored holdings path, the bounded tanh head and the differential Sharpe reward leave concentration 0.010 and turnover 0.001. The outline is a book that barely moves.

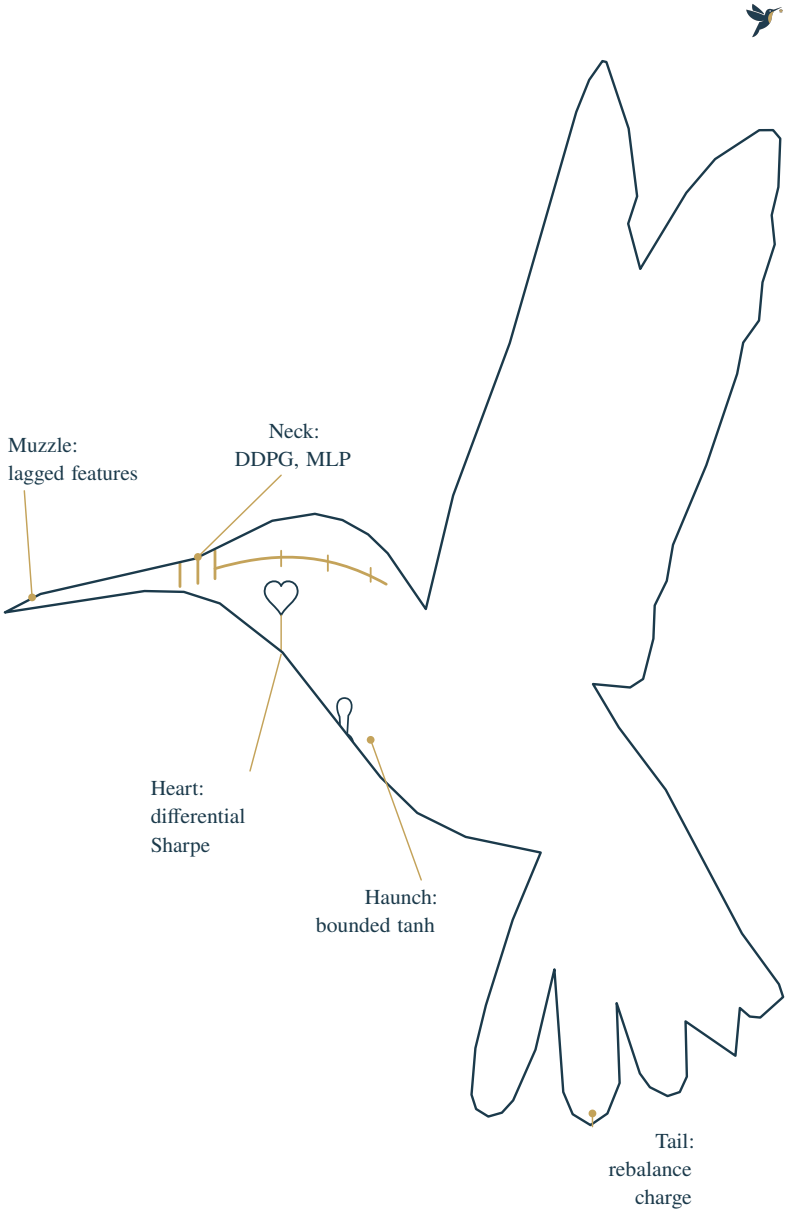


Figure 14. Tactical rotator (Hummingbird). Turnover 0.001, a book that barely moves.

The Hummingbird's turnover is 0.001. Concentration is 0.010, so on 100 names the book sits on the equal-weight floor, and the largest weight is 1.0 percent. Moreira and Muir scale equity down when recent volatility is high (Moreira and Muir, 2017). A desk that follows that rule changes the mix when the regime changes and barely trades inside a regime. This book is DDPG with a bounded tanh head and a differential Sharpe reward. Its market loading is −0.06. The stored path Sharpe is −0.179, so the quiet habit was not paid over the path.

6.12 Contrarian

On the stored holdings path, the softmax head and the mark-to-market reward leave concentration 0.007 and turnover 0.006. The fox is that quiet spread, not a story about cunning.

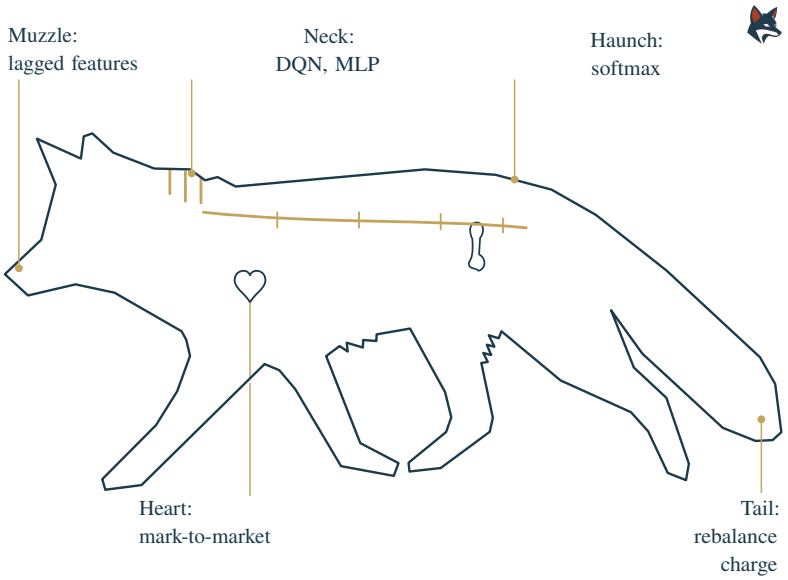


Figure 15. Contrarian (Fox). Softmax head, turnover 0.006, a book spread out and quiet.

The Fox's largest weight is 0.73 percent. Jegadeesh documents a negative serial correlation in monthly stock returns, and a reversal desk buys recent losers and sells recent winners (Jegadeesh, 1990). This book is a DQN with a softmax head and a mark-to-market reward on 200 names. Concentration is 0.007 and one-way turnover is 0.006, so the book stays spread out and barely moves. Its market loading is 0.02, so the path is not a bet on the market. The stored path Sharpe is −0.245. The training file carried no contrarian name. The holdings filed one.

6.13 Generalist

On the stored sheet, concentration is 0.010 and one-way turnover is 0.025. The leading score does not clear 0.25. The gap versus equal weight on this sheet is below one quarter, so the census screen withholds the name. Turnover is 0.025, so the charge on this book stays small.

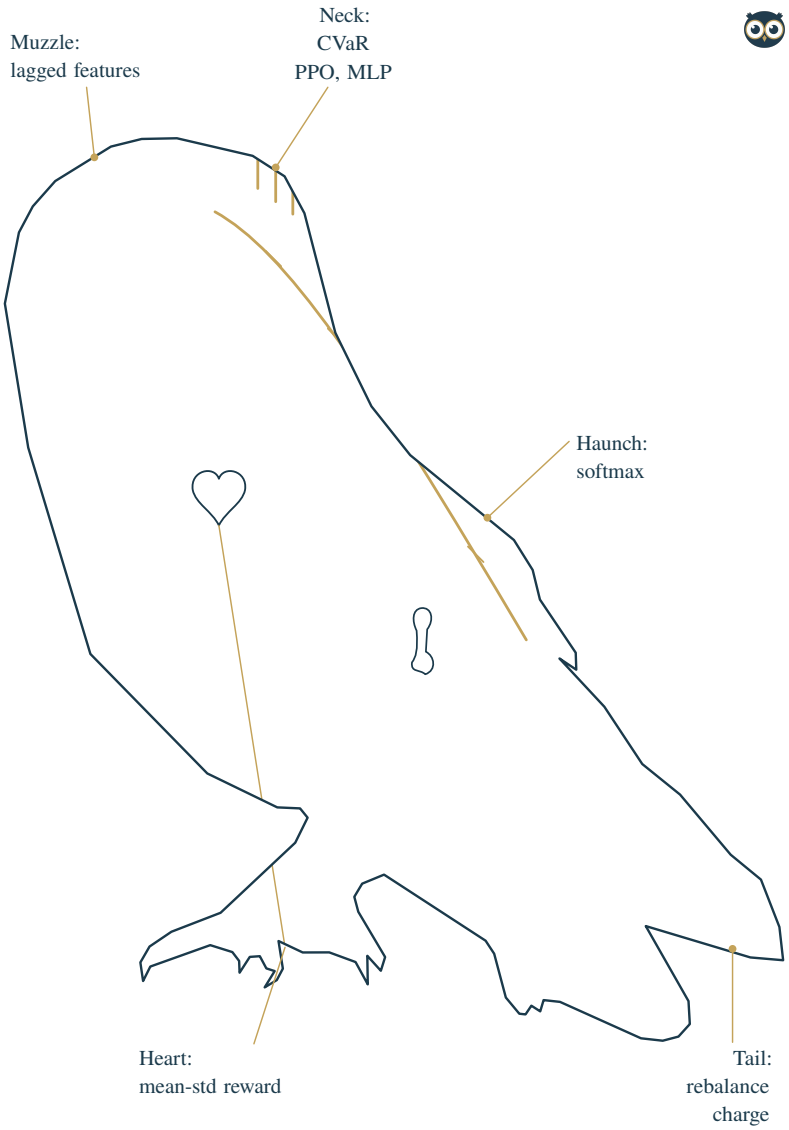


Figure 16. Generalist (Owl). The residual name. Turnover 0.025.

The Owl is filed when no score leads by 0.25. The trainer was not asked to become that name. A multi-strategy desk in that case keeps the book and does not force a style onto a dead heat (Sharpe, 1992). The peer is a blend of the other five rules, and equal weight when no mandate wins.

6.14 One day in the life

Figure 17 is the highest-turnover day in 2018 and 2019 on which the previous book was already invested. The five largest names that day are NRG Energy, Newmont, Republic Services, Expedia, and Everest Group. Gross return is 91.9 basis points, cost is 11.8 basis points, and net reward is 80.1 basis points. On 26 March 2019 the Cheetah bought five names from a weight of zero. The same five firms occupied those slots on 25 March, so the change is a trade: NRG Energy 0.282, Newmont 0.218, Republic Services 0.179, Expedia 0.174, and Everest Group 0.148. Six days earlier the Federal Open Market Committee left the funds target at 2-1/4 to 2-1/2 percent and said it would be patient (Board of Governors of the Federal Reserve System, 2019). The S&P 500 closed at 2798.36 on 25 March and at 2818.46 on 26 March. A momentum desk buys a name whose own past return is positive (Moskowitz, Ooi, and Pedersen, 2012). Expedia’s twelve-minus-one return that day is 2.762, and Everest Group’s is 0.152. NRG Energy, Newmont, and Republic Services, at −0.181, −0.075, and −0.310, were not winners. The desk would add Expedia. The book also bought the three losers. Gross

that day is 91.9 basis points, cost is 11.8, and net is 80.1. Over the stored path the momentum-factor loading is 0.02 and the market loading is 0.87, so the concentration is not a momentum-factor bet. The training file asked for a risk manager. The holdings filed a trend follower. That day's net is not the path, and the path is not offered as a beat of equal weight. Panel 1 shows lagged twelve-minus-one-month momentum for the ten largest weights, labelled by ticker and sector. Panel 2 compares previous and new weights. Panel 3 lists the trades. Panel 4 walks from gross return through cost to net reward.

7 Mechanism atlas: what produces each style

Each personality in Section 6 has its own configuration. Sparsemax can place exact zeros, softmax cannot, and bounded tanh clips the scores and then renormalises. The growth line on each sheet is that book's path with the highest compounded net return. Where that path does not store weights, the holdings panels stay on the path that does. That path lets the sheet be read, and the personalities are not ordered by its return. The last panel is the share of the book in the five largest names. This section reverse-maps each personality. Each subsection states the score and the human peer.

7.1 Trend follower

This subsection is one trend-follower book. The census counts 11 trend-follower books among 179. A trend follower bets that recent winners keep winning. The peer is twelve-minus-one-month momentum (Moskowitz, Ooi, and Pedersen, 2012). The frozen scorecard awards

$$S_{\text{trend}} = 0.5\, z(\text{residual trend}) + 0.2\, z(\text{autocorrelation}) + 0.2\, z(\text{holding period}) - 0.1\, z(\text{reversal}).$$

The book is a sparsemax head with a squared-error hedging reward, and the score loads on residual trend, persistence, and a long holding period, and it penalises reversal. The reward pays for staying close to a trend target and penalises the miss. The sparsemax head then keeps the names whose scores survive the projection onto the simplex and sets the rest to zero (Martins and Astudillo, 2016). That is why a few names can carry the book, and why the last panel, the share in the five largest names, is the right picture of this head. The peer is the same habit written as a published rule: winners over the past year, skipping the most recent month (Moskowitz, Ooi, and Pedersen, 2012). The residual-trend term in the score is that habit after the equal-weight book has been taken out. The update sees the hedge only after the charge, so a winner that does not survive the charge is not the name the next book is paid to keep.

Read the sheet from the top left. The growth panel is the path with the highest compounded net return, set against the momentum peer and against equal weight. The rolling panel is that same path. The holdings panel is the path that stores weights: concentration 0.58 and turnover 0.27, and each row is a firm. The sector stack is the GICS proxy of those weights. Turnover and cumulative cost use the configured charge. The scorecard is the frozen reading that filed the trend-follower name, loading on residual trend, persistence, and a long holding period, and penalising reversal.

7.2 Contrarian

This subsection is one contrarian book. The census counts 32 contrarian books among 179. A contrarian buys recent losers. The peer is one-month reversal (Jegadeesh, 1990).

$$S_{\text{contrarian}} = 0.5\, z(\text{reversal}) + 0.3\, z(\text{turnover}) - 0.2\, z(\text{residual trend}).$$

The book is a full-calendar DQN book with a softmax head and a mark-to-market reward. The reward on this sheet is mark-to-market profit after the charge. That is the outcome the update scores. DQN scores proposed books on a replay of past days and then moves the mapping. This sheet is one discrete book. It is not evidence for the actor-critic books, which learn a policy and a value together (Hilpisch, 2024). The softmax head keeps a positive weight on every name, which is why this sheet's concentration is 0.007 and the share in the five largest names stays low. The score still files a contrarian because the holdings load on reversal and on turnover, and they penalise residual trend. One-month reversal is the published form of that habit (Jegadeesh, 1990). Eight of these books use the short calendar in Section 5.3. They stay in the holdings census and out of any performance sentence.

Read the sheet from the top left. The growth panel is the path with the highest compounded net return, set against one-month reversal. The holdings panel is the path that stores weights: concentration 0.007 and turnover 0.006. The head is softmax, so the book does not place exact zeros, and the weights stay spread out. The score loads on reversal and on turnover, and it penalises residual trend. Eight books in this personality use the short calendar and stay out of any performance sentence.

7.3 Risk manager

This subsection is one risk-manager book. The census counts 18 risk-manager books among 179. A risk manager seeks a quieter book. The peer is shrunk minimum variance (Clarke, Silva, and Thorley, 2006).

$$S_{\text{risk}} = 0.4\, z(\text{defensive tilt}) + 0.3\, z(\text{minus downside capture}) + 0.2\, z(\text{minus largest weight}) + 0.1\, z(\text{defensive response to volatility}).$$

The book is an LSTM torso and a sparsemax head with a mean-standard-deviation reward, and the score loads on defensive tilt, minus downside capture, minus the largest weight, and the defensive response to volatility. The reward pays for average profit and penalises the recent scatter of that profit, so a quieter book scores higher than a louder one with the same average. The LSTM torso can carry a state across days. The head still decides the weights, and the personality is filed from those weights. Sparsemax can set names to zero. On this sheet the turnover is 0.010, so the zeros, once chosen, are held. The peer is a shrunk minimum-variance book, the published form of the same preference for a quieter mix (Clarke, Silva, and Thorley, 2006).

Read the sheet from the top left. The growth panel is the path with the highest compounded net return, set against shrunk minimum variance. The holdings panel is the path that stores weights: concentration 0.46 and turnover 0.010. The low turnover is the quiet book this score is built to recognise. The torso is an LSTM and the head is sparsemax. The score loads on defensive tilt, minus downside capture, minus the largest weight, and the defensive response to volatility.

7.4 Speculator

This subsection is one speculator book. The census counts 21 speculator books among 179. A speculator concentrates in high-lottery names. The peer is the MAX lottery sort (Bali, Cakici, and Whitelaw, 2011). 21

books take this name.

$$S_{\text{speculator}} = 0.4\,z(\text{lottery tilt}) + 0.3\,z(\text{concentration}) + 0.2\,z(\text{largest weight}) + 0.1\,z(\text{upside capture}).$$

The pictured sheet is a softmax hold with concentration 0.99. The census files that hold as a generalist. The peer for the speculator name is the MAX sort (Bali, Cakici, and Whitelaw, 2011). The reward is profit after the charge. It does not pay for being spread out. Softmax never places an exact zero, and this book's concentration is still 0.99, because the scores that reach the head are already peaked. The last panel is that share in the five largest names. Every name can keep a small weight, and the money can still sit in a few of them. The MAX peer buys the names with the largest recent daily move. That is the published form of the lottery habit the score is built to detect (Bali, Cakici, and Whitelaw, 2011).

Read the sheet from the top left. The growth panel is the path with the highest compounded net return, set against the MAX sort. The holdings panel is the path that stores weights: concentration 0.99 and turnover 0.002. The head is softmax, so that concentration sits in the scores the head receives. The score loads on lottery tilt, concentration, the largest weight, and upside capture.

7.5 Tactical rotator

This subsection is one tactical-rotator book. The census counts 19 tactical-rotator books among 179. A tactical rotator moves the book as the mix of styles changes. The peer is volatility-managed equity (Moreira and Muir, 2017).

$$S_{\text{rotator}} = 0.5\,z(\text{rotation}) + 0.3\,z(\text{tilt variance across regimes}) - 0.2\,z(\text{tilt variance within a regime}).$$

The book is a bounded tanh head with a differential Sharpe reward, and the score loads on rotation and on tilt variance across regimes, and it penalises tilt variance within a regime. The reward, following Moody and Saffell (2001), pays for an improvement in the recent ratio of average profit to its scatter, and the charge is already inside that profit. Bounded tanh clips each score and then renormalises, so the book can lean without placing exact zeros and without letting one score take the whole fund. That is why concentration is 0.010 and turnover is 0.001: the book changes its mix across regimes and barely changes it inside a regime. Volatility-managed equity is the published cousin of that habit (Moreira and Muir, 2017).

Read the sheet from the top left. On this book the path with the highest compounded net return is also the path that stores the weights. Concentration is 0.010 and turnover is 0.001. The head is bounded tanh, so scores are clipped and then renormalised. The score loads on rotation and on tilt variance across regimes, and it penalises tilt variance within a regime. The peer is volatility-managed equity.

7.6 Generalist

This subsection is one generalist book. The census counts 78 generalist books among 179. A generalist is a book whose winning score does not lead by 0.25. The peer is a blend of the other five rules. The label is the filing when no single mandate fits. The leading score does not lead by 0.25, which is the rule that files the generalist label. The filing does not say the book is diffuse. It says no one score leads. On this sheet the concentration is 0.010 and the turnover is 0.025. The reward is mean profit against recent scatter, the same family as the risk-manager reward, and the torso is an MLP, which keeps no hidden state past today's lagged

Table 6. Personality card for one trend-follower book. The animal nickname sits under the name once.

Field	Entry
Label	Trend follower (Cheetah)
Gap	1.00
Active share	About 0.94; the screen is 12.5 percent
Training tuple	PPO, MLP torso, sparsemax head, squared-error hedging reward, 100 names
Top holdings	On 26 March 2019: NRG Energy, Newmont, Republic Services, Expedia, Everest Group
Drift triggers	Label change, or 63-day active share below 12.5 percent

features. Equal weight is the peer because a blend of the other five rules is what the filing means when no mandate wins.

Read the sheet from the top left. The growth panel is the path with the highest compounded net return, and that path stores the weights. The leading score does not clear 0.25. The sheet sits below the closet-index screen, so the census does not count it. Concentration is 0.010 and turnover is 0.025. The peer on this sheet is equal weight, because no single mandate fits.

7.7 Design choices and predicted behaviour

Table 5 records how the allocation head changes concentration and turnover. Appendix F lists the most common rewards and heads among the filed books.

8 What a personality costs

The charge is five basis points times the change in weights. The same charge is inside the training step and inside the evaluation. A book that barely changes its weights therefore pays little. A book that replaces the book pays a great deal. On the holdings paths the measured cost runs from 6.2 to 129 basis points of turnover, with a median of 10.0. That range is the turnover of the books, not a second fee schedule. Implementation shortfall against an arrival price is a different measurement (Grinold and Kahn, 2020).

The trend-follower sheet turns over 0.27 of the book on the path that stores weights, so the charge is a regular part of the step. The contrarian sheet turns over 0.006, and the tactical rotator turns over 0.001. Those two books can look cheap because the head, softmax or bounded tanh, keeps weight on many names and moves them only a little. The risk-manager sheet turns over 0.010. The sparsemax head can set names to zero, and this book still holds them for a long stretch, so the defensive score and the low turnover arrive together. The speculator sheet turns over 0.002 and still places 0.99 of the concentration measure on a few names. The cost stays small because the book does not keep replacing those names. The generalist sheet turns over 0.025. The leading score does not clear 0.25, the sheet sits below the closet-index screen, and the charge stays small because the weights barely move.

The configured charge in Section 6 is this same charge. It does not choose the personality. The personality is filed from the holdings after the charge has already been taken.

9 From personality to approval file

The card records the label, the runner-up, the gap, and the screen.

10 Limitations

The walk-through book is one seed. The head table uses seeds 0 through 4. Eight contrarian books use the short calendar in Section 5.3. The vendor data are licensed and are not redistributed. Three 400-name books are outside the hundred-slot join. Carry and illiquidity are not in the residual tags. DeMiguel, Garlappi, and Uppal (2009) is a boundary: optimised rules often fail to beat one-over-N, and that contest is not this paper's finding. That rare match is the designed name against the filed personality. The defensive tilt in the risk-manager score can sit on a published low-risk factor. The sparse head's exact zeros can produce the concentration the score then reads as a speculator. The correlation-cluster slot rule is part of every label in the census. The census does not separate a change of who occupies a slot from a change of size. The charge is a turnover rate, not a venue fill.

11 Conclusion

The question was whether a reinforcement-learning equity book can be summarised by a pre-registered trading personality that a reader can verify from the holdings alone. The screen uses holdings only. It withholds the name when the mean absolute gap versus equal weight is below one quarter of the fund. On the 179 books that clear the screen, the scorecard files 11 trend followers, 32 contrarians, 18 risk managers, 21 speculators, 19 tactical rotators, and 78 generalists. The card records the name, the runner-up, and the gap. A string in the configuration file does not supply that name.

The same book is then read as one body. The muzzle is the lagged state, the neck is the network torso, the collar is the map onto the simplex, the heart is the reward, the haunch is the allocation head, the tail is the charge used in training and in evaluation, and the spine is the learning update. Sparsemax can leave a weight at exactly zero. Softmax cannot. Bounded tanh clips the scores and then renormalises. On the 60-book PPO MLP factorial, the reward inside a fixed head barely moves turnover, while the three heads do not sit on top of one another. The personality is not the head. The personality is the name filed from the holdings.

No tested book is offered as a beat of a same-cost benchmark. The three 400-name books sit outside the census because the style matrix does not cover them. Their stored paths are a reading of torso and cadence, not a second census.

11.1 Open problems this framework makes tractable

A second seed, and a drift monitor that writes the same personality card, are open.

11.2 Reproducibility

Vendor microdata are licensed and are not redistributed. A public note shows how to stand up a book of this kind. The stored results that support this paper are in this folder. The implementation is retained by the author and is not part of that note. Results are backtested and do not constitute investment advice.

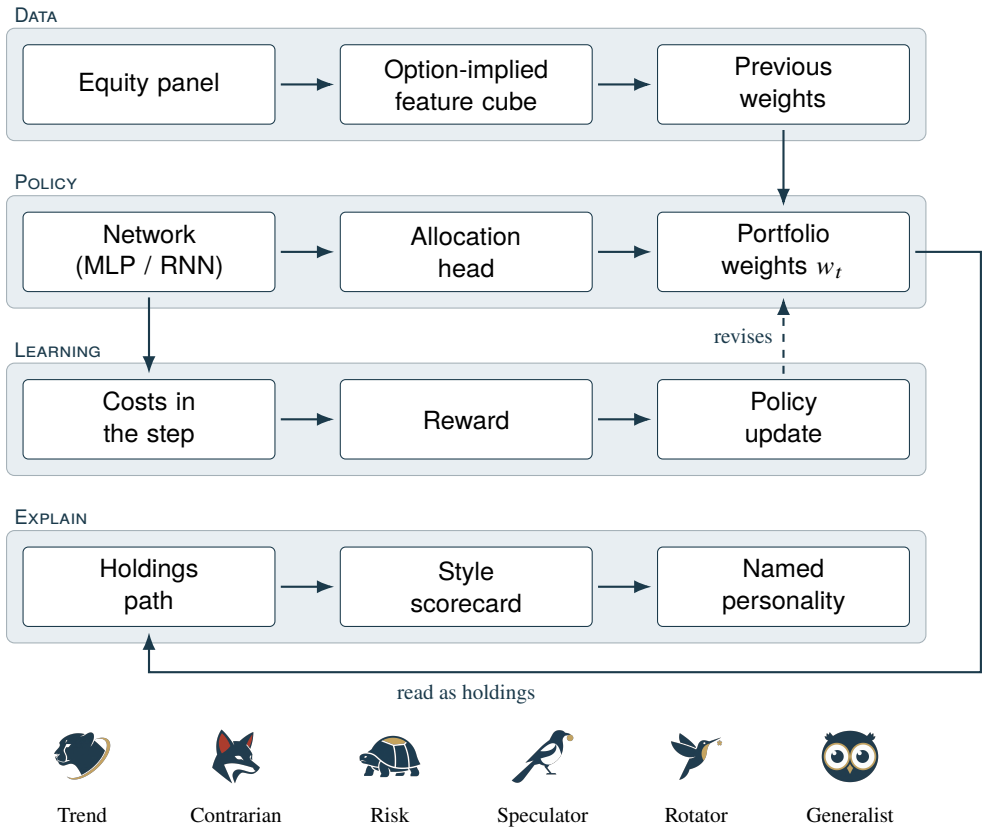


Figure 2. Architecture of the study as four lanes. Data feeds a policy that writes simplex weights; learning updates the policy with costs inside the step; explanation scores the holdings path into a named personality. Section 6 draws one body for each personality; Section 7 reverse-maps each populated style.

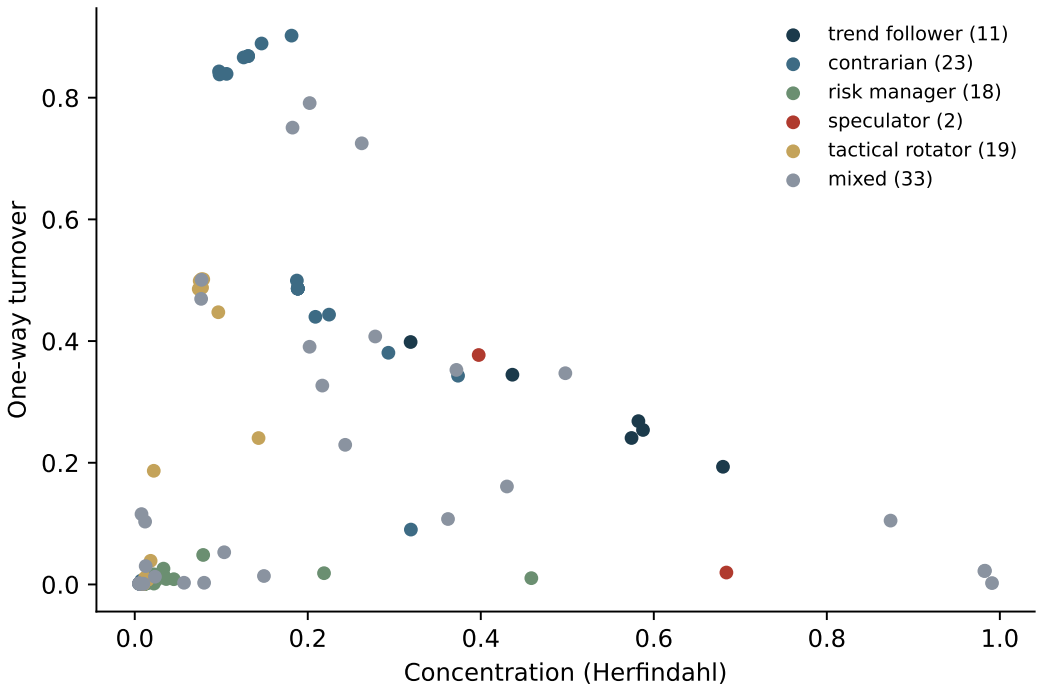


Figure 3. Concentration against one-way turnover for the 106 books that already carried a designed name. The two filed speculators sit toward high concentration. The risk managers sit at low concentration and low turnover.

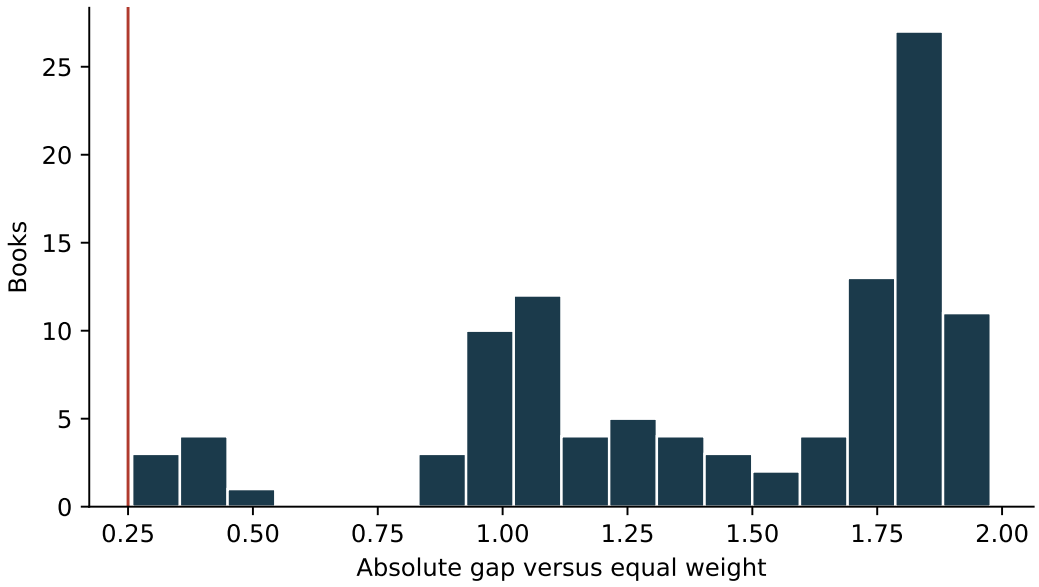


Figure 4. Absolute gap versus equal weight for the same 106 books. The vertical line is the closet-index screen at one quarter. Every book in this figure clears it.

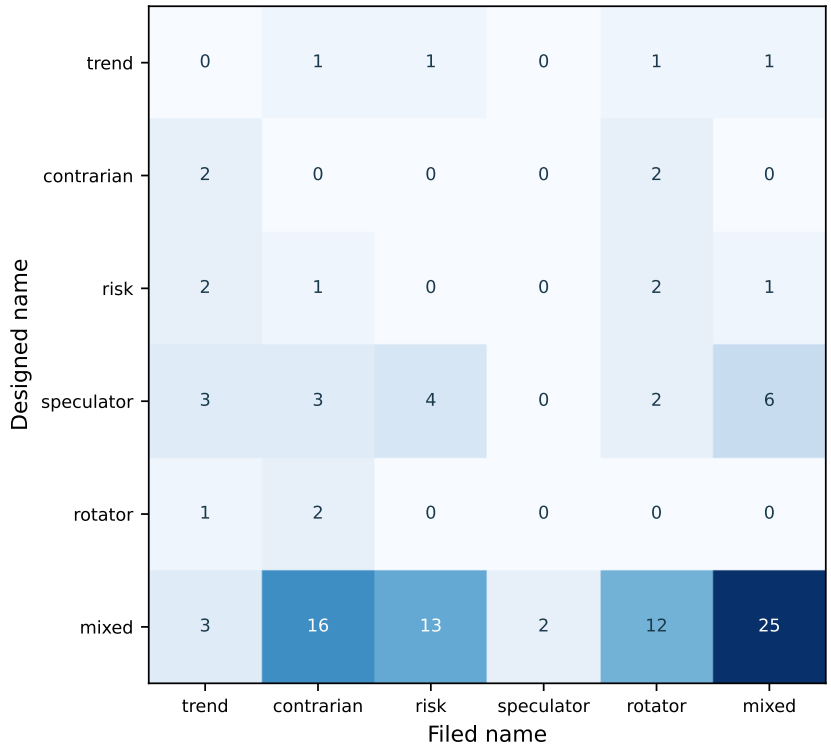


Figure 5. Designed name against the name the scorecard filed, on the 106 books. A cell is a count. Sharing a label is more common than the strict match of 7, and a mismatch is not a bad result.

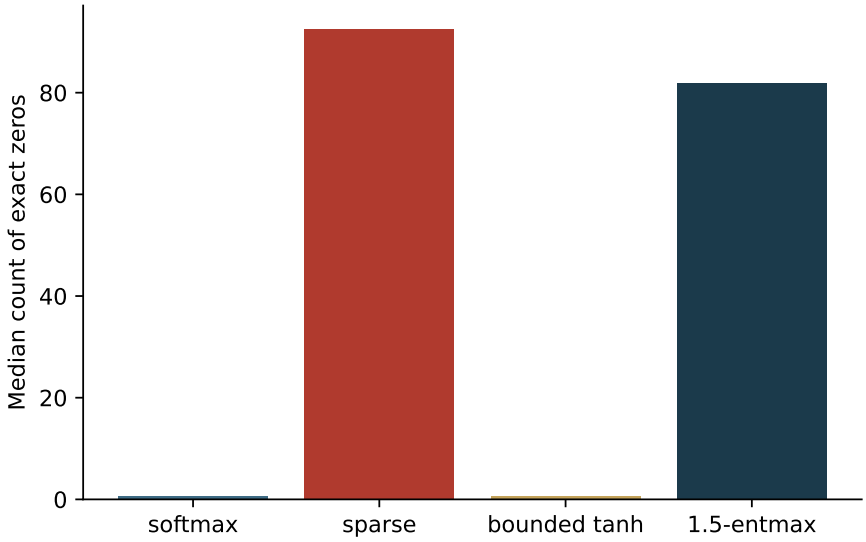


Figure 8. Median count of exact zeros in the stored weights, by allocation head. The count is the mean over every tenth day of the stored path, then the median across books with that head.

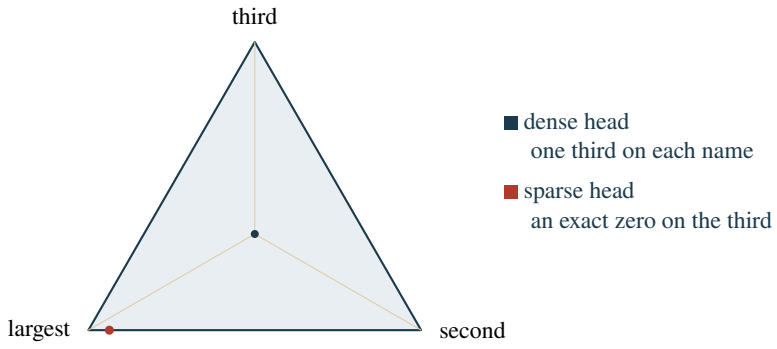


Figure 9. The three largest weights on 3 July 2019, renormalized to the simplex. The dense head sits near one third on each name. The sparse head puts an exact zero on the third name and still sums to one.

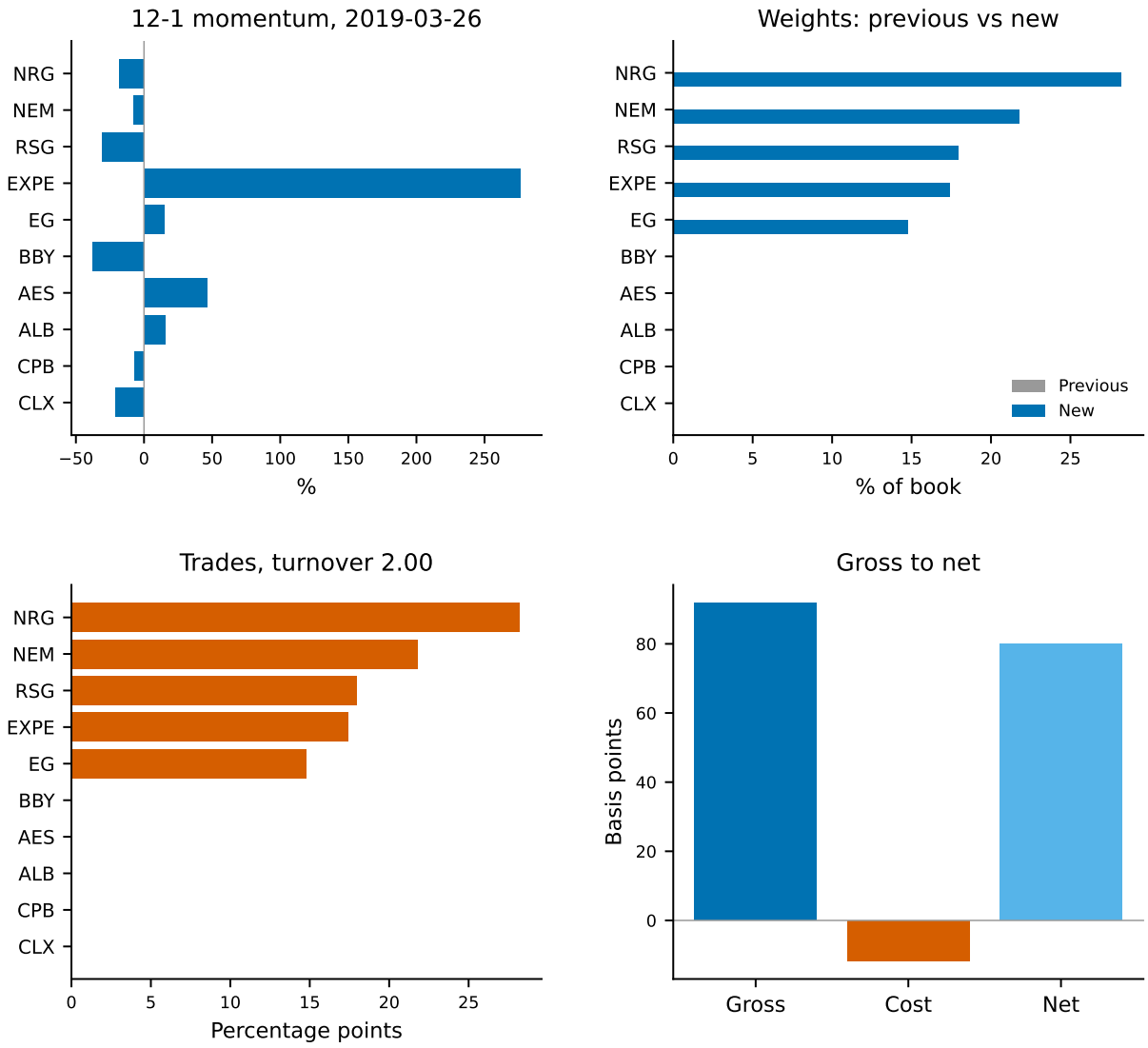


Figure 17. One trend-follower book on 26 March 2019: lagged momentum for the largest holdings, previous versus new weights, the trade list, and gross, cost, and net.

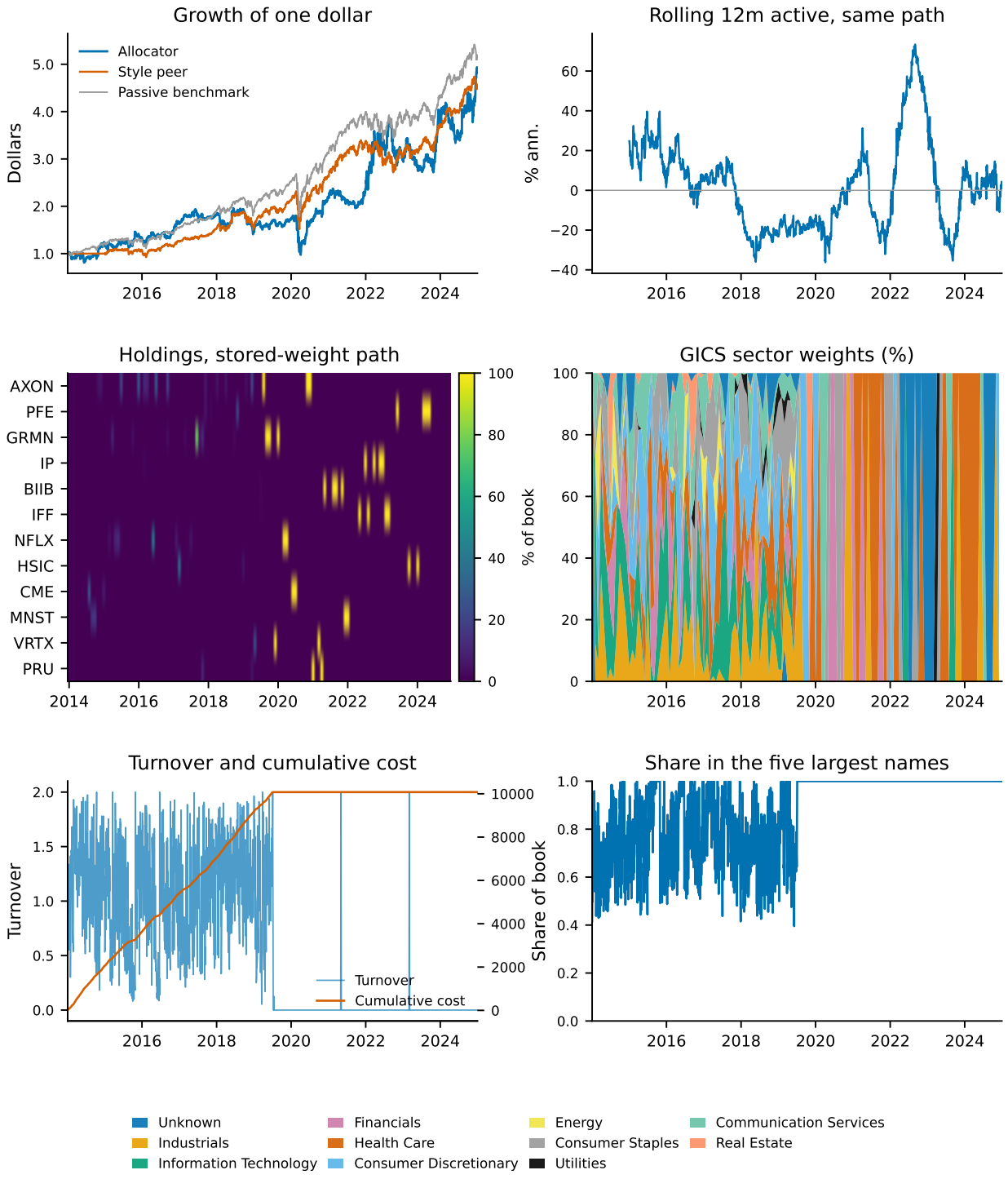


Figure 18. One trend-follower book: growth of one dollar, rolling active return versus the momentum peer, named holdings, sector stack, turnover and cost, and the share in the five largest names.

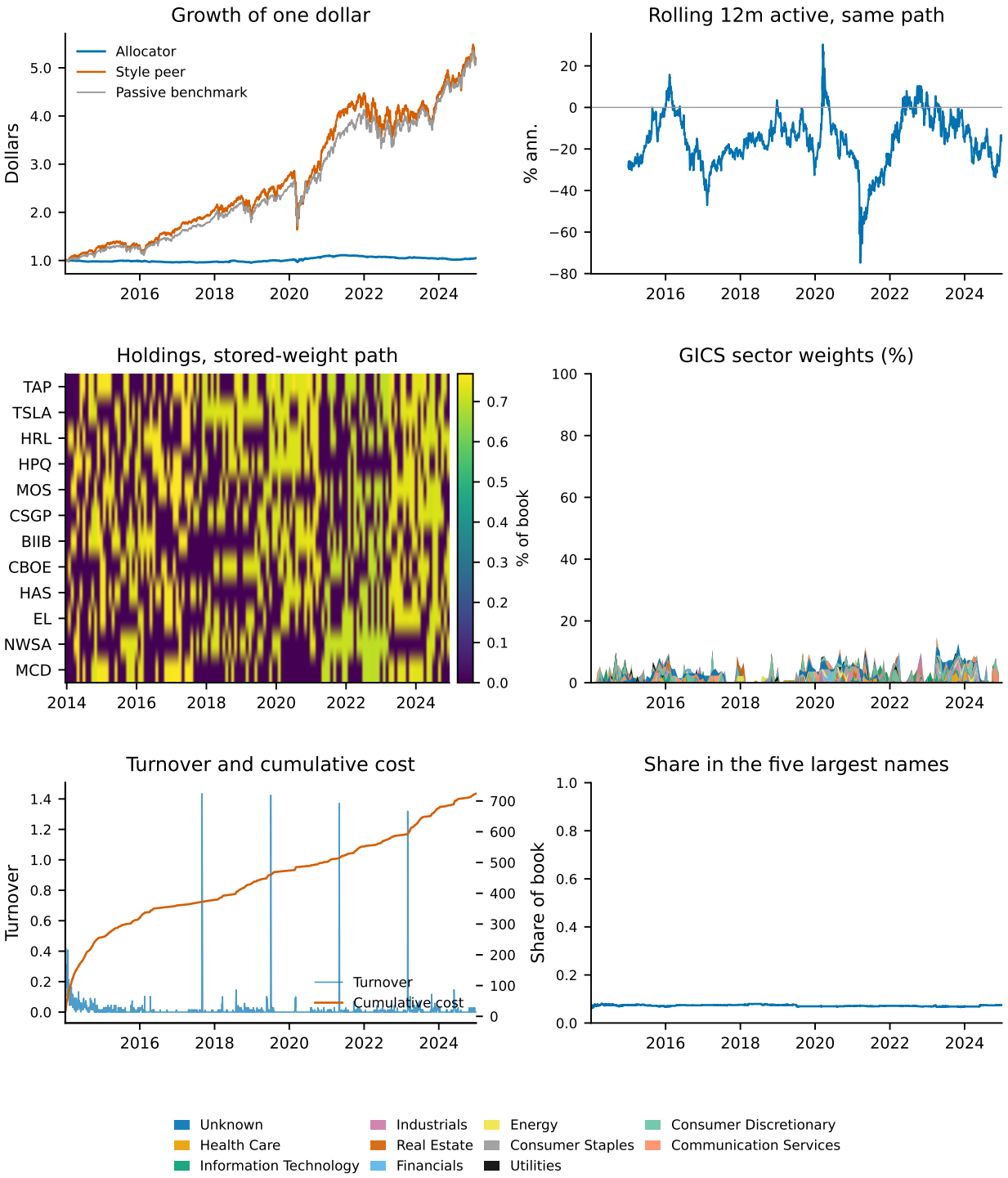


Figure 19. One contrarian book: growth of one dollar, rolling active return versus the reversal peer, named holdings, sector stack, turnover and cost, and the share in the five largest names.

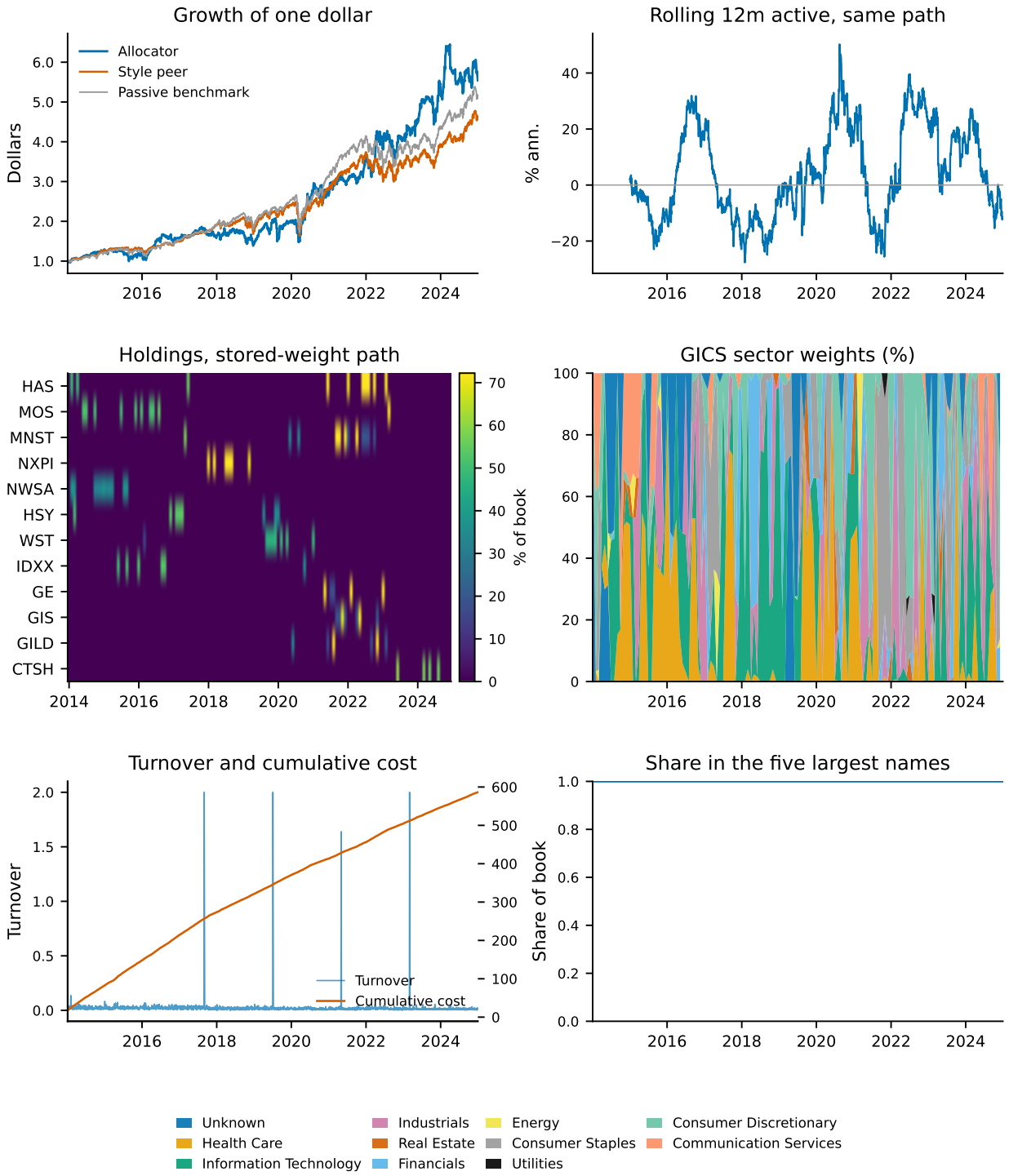


Figure 20. One risk-manager book: growth of one dollar, rolling active return versus the shrunk minimum-variance peer, named holdings, sector stack, turnover and cost, and the share in the five largest names.

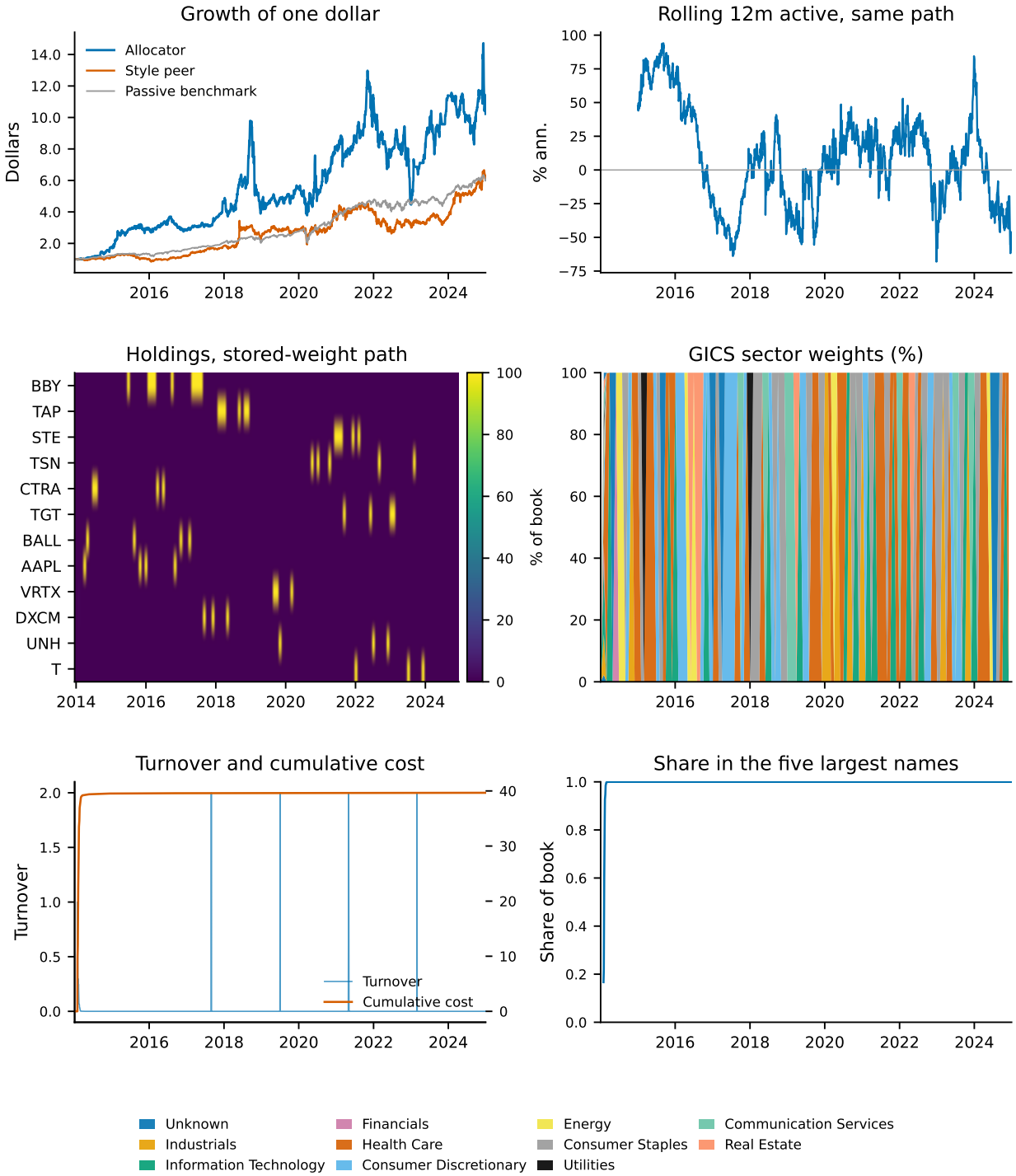


Figure 21. One speculator book: growth of one dollar, rolling active return versus the MAX peer, named holdings, sector stack, turnover and cost, and the share in the five largest names.

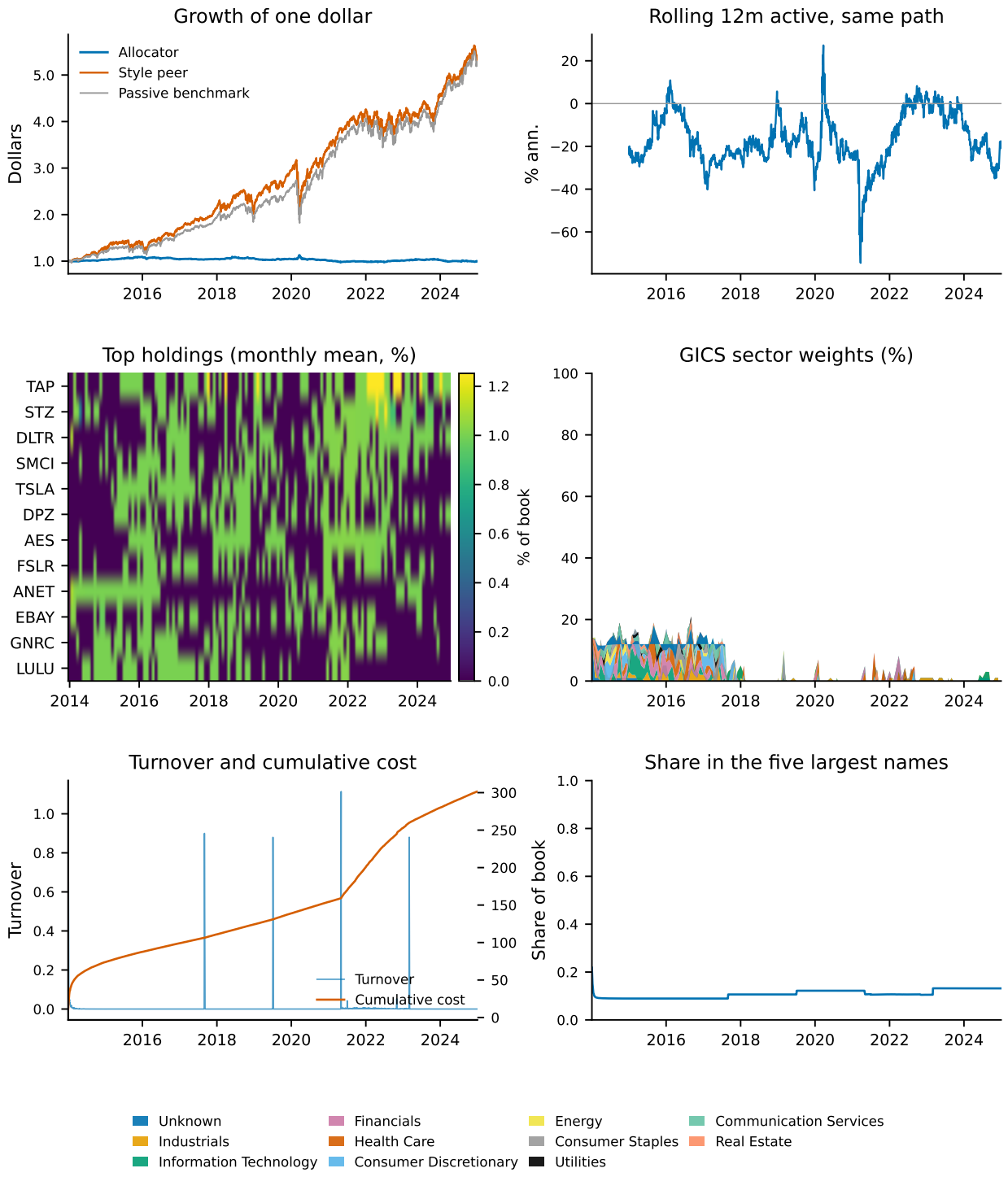


Figure 22. One tactical-rotator book: growth of one dollar, rolling active return versus the volatility-managed peer, named holdings, sector stack, turnover and cost, and the share in the five largest names.

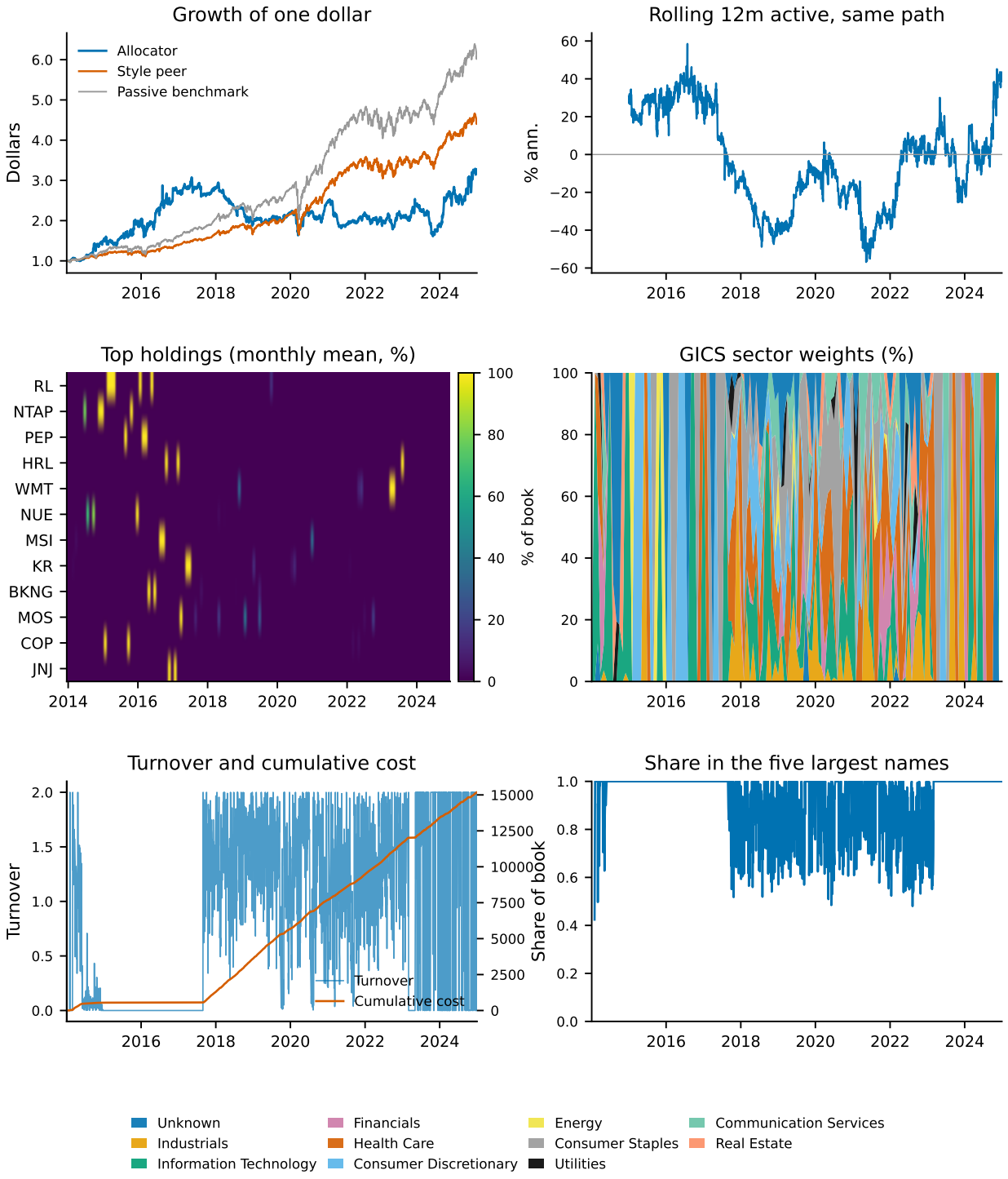


Figure 23. One generalist book: growth of one dollar, rolling active return versus equal weight, named holdings, sector stack, turnover and cost, and the share in the five largest names.

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Appendix

A Archetype scorecard

Each personality except the generalist is a weighted sum of holding measures. The weights below were locked before the census. The measure names match the scores in Section 7.

Table 7. Frozen scorecard. A positive weight raises that personality's score when the measure is high. A negative weight lowers it. The generalist has no row: it is the filing when no score leads by 0.25.

Personality	Measure	Weight
Trend follower	Residual trend	0.50
	Autocorrelation	0.20
	Holding period	0.20
	Reversal	−0.10
Contrarian	Reversal	0.50
	Turnover	0.30
	Residual trend	−0.20
Risk manager	Defensive tilt	0.40
	Minus downside capture	0.30
	Minus largest weight	0.20
	Defensive response to volatility	0.10
Speculator	Lottery tilt	0.40
	Concentration	0.30
	Largest weight	0.20
	Upside capture	0.10
Tactical rotator	Rotation	0.50
	Tilt variance across regimes	0.30
	Tilt variance within a regime	−0.20

B Universe firm list

Most frequently held $K = 100$ names by slot-day share, with ticker, name, GICS sector, and industry:

Ticker	Name	GICS sector	Industry	Share (%)
TAP	Molson Coors Beverage Co	Consumer Staples	Beverages	1.43
STZ	Constellation Brands, Inc.	Consumer Staples	Beverages	1.17
NEM	Newmont Corporation	Materials	Metals & Mining	0.92
SMCI	Super Micro Computer Inc	Information Technology	Technology Hardware, Storage & Peripherals	0.89
FSLR	First Solar, Inc.	Information Technology	Semiconductors & Semiconductor Equipment	0.88
DPZ	Domino's Pizza, Inc.	Consumer Discretionary	Hotels, Restaurants & Leisure	0.88
ERIE	Erie Indemnity Co	Financials	Insurance	0.86
HAS	Hasbro, Inc.	Consumer Discretionary	Leisure Products	0.85
AXON	Axon Enterprise, Inc.	Industrials	Aerospace & Defense	0.85
TSLA	Tesla Inc	Consumer Discretionary	Automobiles	0.83
TMUS	T-Mobile US Inc	Communication Services	Wireless Telecommunication Services	0.79
EA	Electronic Arts Inc	Communication Services	Entertainment	0.78
GEN	Gen Digital Inc.	Information Technology	Software	0.77

Ticker	Name	GICS sector	Industry	Share (%)
DLTR	Dollar Tree Inc	Consumer Staples	Consumer Staples Distribution & Retail	0.77
DISCB	Warner Bros. Discovery, Inc.	Communication Services	Entertainment	0.75
TSN	Tyson Foods Inc	Consumer Staples	Food Products	0.74
T	AT&T Inc	Communication Services	Diversified Telecommunication Services	0.74
DXCM	Dexcom Inc	Health Care	Health Care Equipment & Supplies	0.72
AKAM	Akamai Technologies Inc	Information Technology	IT Services	0.70
KR	Kroger Co	Consumer Staples	Consumer Staples Distribution & Retail	0.70
WBD	Warner Bros. Discovery, Inc.	Communication Services	Entertainment	0.70
EBAY	eBay Inc.	Consumer Discretionary	Broadline Retail	0.69
DVA	DaVita Inc	Health Care	Health Care Providers & Services	0.66
GNRC	Generac Holdings Inc.	Industrials	Electrical Equipment	0.66
MOS	Mosaic Co	Materials	Chemicals	0.65
DECK	Deckers Outdoor Corporation	Consumer Discretionary	Textiles, Apparel & Luxury Goods	0.65
CMCSA	Comcast Corporation	Communication Services	Diversified Telecommunication Services	0.64
CMG	Chipotle Mexican Grill, Inc.	Consumer Discretionary	Hotels, Restaurants & Leisure	0.64
ULTA	Ulta Beauty Inc	Consumer Discretionary	Specialty Retail	0.63
MNST	Monster Beverage Corp	Consumer Staples	Beverages	0.62
ADM	Archer-Daniels-Midland Co	Consumer Staples	Food Products	0.61
RMD	ResMed Inc.	Health Care	Health Care Equipment & Supplies	0.61
BALL	Ball Corp	Materials	Containers & Packaging	0.60
NFLX	Netflix, Inc.	Communication Services	Entertainment	0.60
BMY	Bristol-Myers Squibb Company	Health Care	Pharmaceuticals	0.60
LH	Labcorp Holdings Inc.	Health Care	Health Care Providers & Services	0.60
NRG	NRG Energy Inc	Utilities	Electric Utilities	0.58
LULU	lululemon athletica inc.	Consumer Discretionary	Textiles, Apparel & Luxury Goods	0.58
COST	Costco Wholesale Corp	Consumer Staples	Consumer Staples Distribution & Retail	0.57
BIIB	Biogen Inc.	Health Care	Biotechnology	0.57

C Books outside the hundred-slot join

The style matrix does not cover a 400-name book, which is why these three are here and not in the census. They are a check on two design choices the census cannot see: the torso, and whether the book rebalances every day or once a month. One book uses an MLP torso. One uses a transformer torso. One uses a transformer torso and rebalances monthly.

On the stored holdings path, the MLP book and the transformer book are the same stored holdings. Mean Herfindahl is 0.004 and mean one-way turnover is 0.022. The monthly book has mean Herfindahl 0.009 and mean one-way turnover 0.013. Equal weight on 400 names has Herfindahl 0.0025, so these books stay spread out. The pictured trend follower, by contrast, has turnover 0.27 and concentration 0.58. On every tenth day the share of names at exact zero is 0.10 for the MLP and the transformer, and 0.29 for the monthly book.

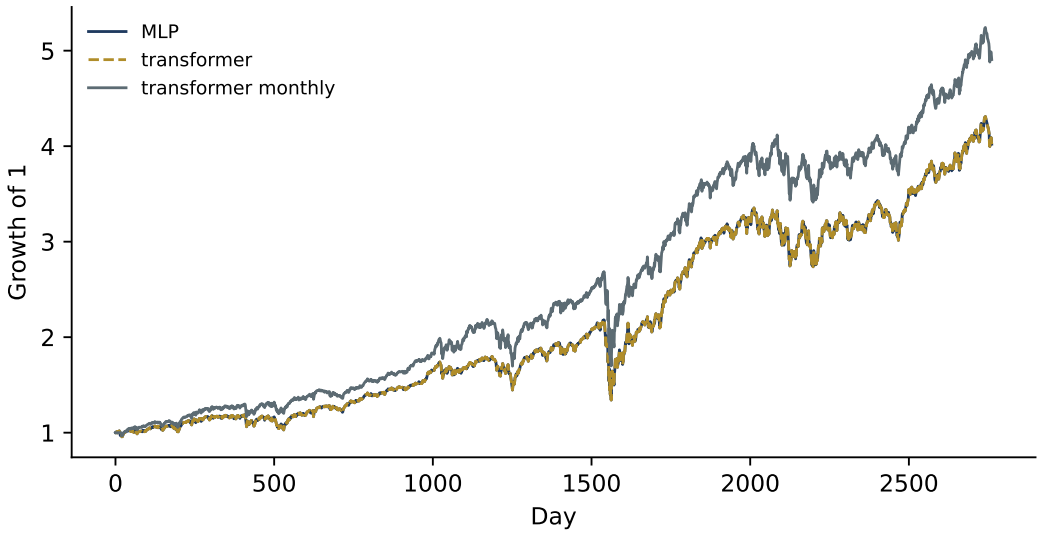


Figure 24. Stored after-cost path for the three 400-name books. The MLP series and the transformer series are the same stored path, so the dashed line sits on the solid one. The lines are not a ranking and not a claim against a benchmark.

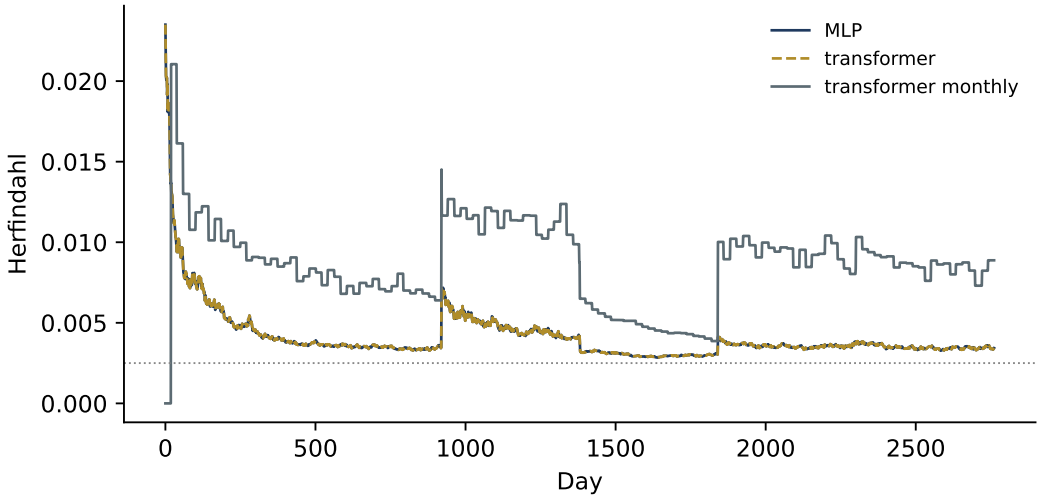


Figure 25. Herfindahl through time on the stored holdings path. The dotted line is equal weight on 400 names. The MLP and transformer series coincide. This is not a ranking.

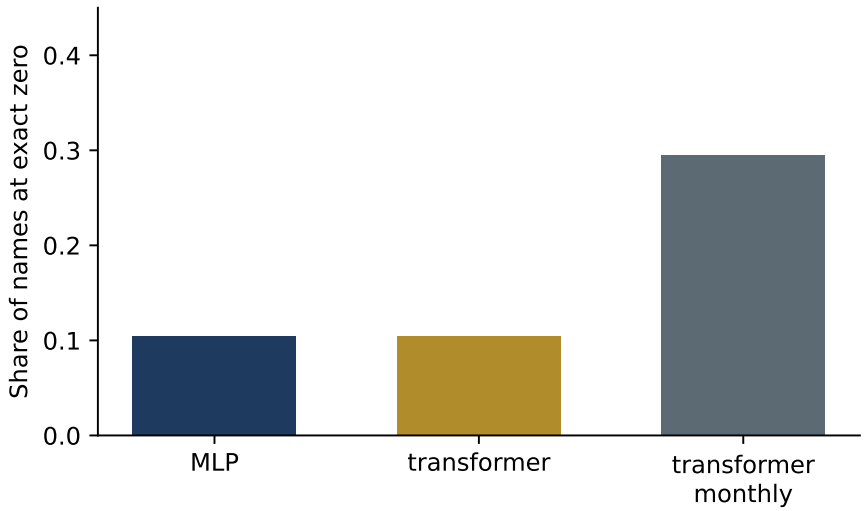


Figure 26. Share of names at exact zero, mean over every tenth day. Set next to the head facts in the body. This is not a new census.

D Processing times

Luxembourg's CSSF tracks processing time from a complete filing to authorisation (Commission de Surveillance du Secteur Financier, 2026). Ireland's Central Bank typically returns first comments within twenty working days. Practitioners budget ten to twelve weeks for a standard file.

E Matched-head twins

Matched-head twins keep their own denominator. They are not the census, and they are not a regression of style on design choices. The median gap in absolute distance from equal weight is 1.673 (Wilcoxon $p = 1.75 \times 10^{-11}$).

Matched sparse vs full-support twins	Value
Number of pairs	58
Median ΔL_1 (sparse − full-support)	1.673
Wilcoxon one-sided p ($\Delta L_1 > 0$)	1.75×10^{-11}

F Design-choice tables

The two tables count filed books by the reward they were trained on, and by the head that wrote the weights. A row is one of the most common choices. The lists do not include every book in the census.

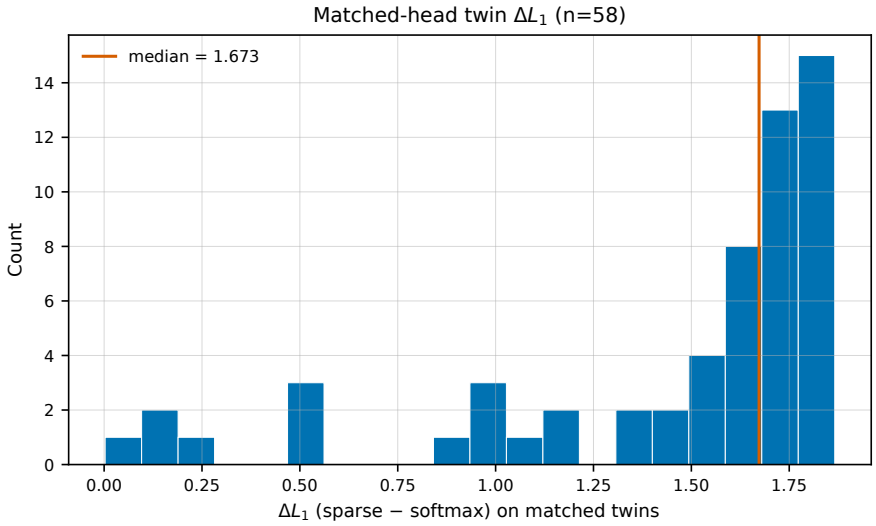


Figure 27. Matched sparse versus full-support twin ΔL_1 distribution.

Table 9. Most common training rewards. The number is how many filed books used that reward. The list is the top of the distribution, not every book in the census.

Personality	Reward	Books
Trend follower	Mark-to-market profit	4
	Mean-variance	3
	Entropic risk	1
	Mean-variance replication	1
Contrarian	Mean-variance	15
	Mark-to-market profit	3
	Asymmetric profit	2
	CVaR	1
Risk manager	Mark-to-market profit	14
	Mean-variance	3
	Differential Sharpe	1
Speculator	Mean-variance	2
Tactical rotator	Differential Sharpe	4
	Asymmetric profit	4
	Mark-to-market profit	4
	Mean-variance	3
Generalist	Mark-to-market profit	14
	Mean-variance	11
	Differential Sharpe	3
	Asymmetric profit	2

Table 10. Most common allocation heads. The number is how many filed books used that head. The list is the top of the distribution, not every book in the census.

Personality	Allocation head	Books
Trend follower	Sparsemax	8
	Softmax	3
Contrarian	Sparsemax	19
	Bounded tanh	3
	Softmax	1
Risk manager	Softmax	9
	Sparsemax	9
Speculator	Sparsemax	2
Tactical rotator	Sparsemax	9
	Bounded tanh	7
	Softmax	2
	Discrete	1
Generalist	Sparsemax	15
	Bounded tanh	10
	Softmax	4
	1.5-entmax	4